USN

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Dec-13/Jan-14

Electronics and Communication Engineering

ANALOG MICROELECTRONIC CIRCUITS

Time: 03 Hours | *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first 3 pages of the answer book only.
2. Answer FIVE full questions from Part B.

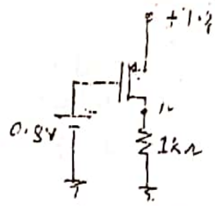
PART-A

1 1.1 A MOSFET, whose $\lambda = 0.02V^{-1}$, carries a drain current $1.1mA$ with $V_{DS} = 5V$ in saturation. If V_{DS} is increased to $10V$, then drain current of the device is 1.2mA 01

1.2 For a MOS transistor, if W/L is increased 16 times while maintaining its I_D constant, its g_m will increase by a factor of 4. 01

1.3 For the circuit shown in fig 1.3, 500 value of W/L places the P-channel device in saturation region.

$I_D = 1mA$.



$Kp' = 100\mu A/V^2$

$Vt = -0.2 V$

$I_D = \frac{1}{2} k' \frac{W}{L} V_{ov}^2$

Fig 1.3 02

1.4 A BJT whose V_A[Early voltage] is $52V$ at 27°C, has an intrinsic gain of 2K.

$\frac{V_A}{V_T}$ $\frac{52}{26}$ 01

1.5 In the circuit of fig 1.5, if the base current of the silicon transistor is $10\mu A$ and its $\beta = 48$, the value of R_C is 10KΩ

$g_m = \sqrt{2 I_D k' \frac{W}{L} 16}$

$A = \frac{V_A}{V_T}$

$= \frac{52}{26} 2K$

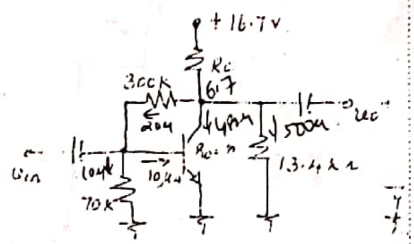
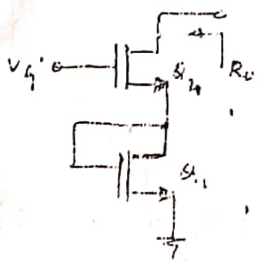


Fig 1.5 02

1.6 If V_{BE} of BJT is $650mV$ at $1\mu A$, then the value of V_{BE} at $25\mu A$ is 0.733 Assume the temperature to be the same. 01

1.7 The output impedance R_0 in the circuit of fig 1.7 is ~~117.0.73~~ K $\boxed{OR}$ 148.047K



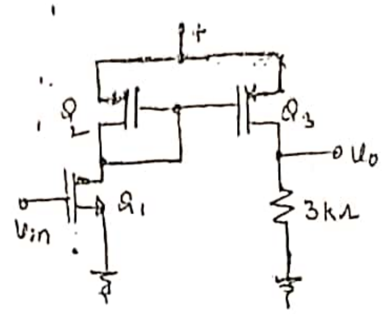
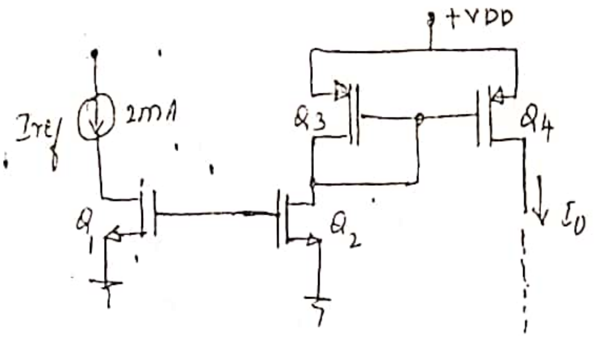
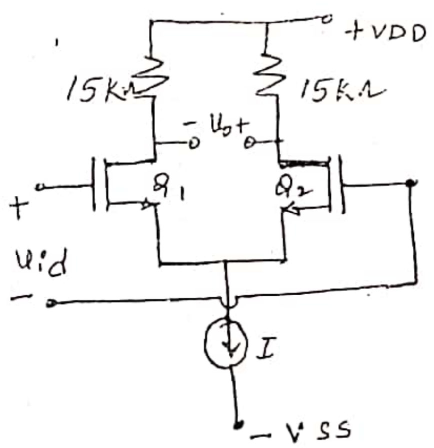
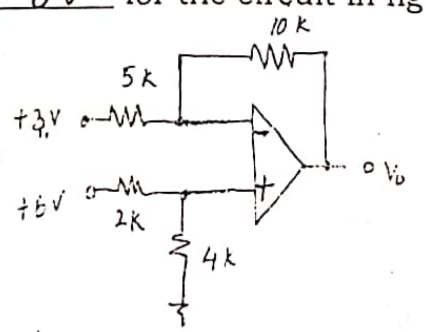
$gm_2 = 4 mA/V$

$r_{o2} = 30K$

$gm_1 = 1mA/V$

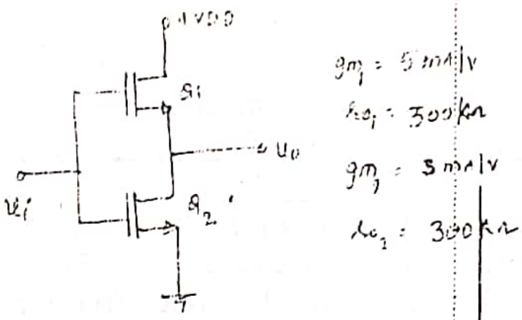
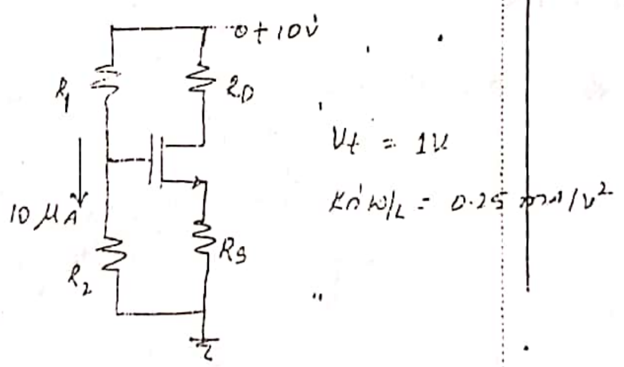
$r_{o1} = 40KΩ$

Fig 1.7 01

1.8	The small signal voltage gain $\frac{V_o}{V_{in}}$ in the circuit of fig 1.8 is 12. Given $g_{m1} = \frac{1mA}{V}, g_{m2} = \frac{2mA}{V}, (W/L)_3 = 4(W/L)_2$  Fig 1.8	01
1.9 2	The current steering circuit of fig 1.9 uses devices having equal channel, but with widths $W_1 = 5\mu m, W_2 = 15\mu m, W_3 = 20\mu m$ and $W_4 = 50\mu m$.The value of I_0= 15 mA  Fig 1.9	01
1.10 3	For the NMOS differential circuit given in fig 1.10, the small signal differential gain, $\frac{v_o}{v_{id}}$= 15. Given $g_{m1} = g_{m2} = 1mA/V$.  Fig 1.10	01
1.11 4	The output voltage, v_0= 6V for the circuit in fig 1.11.  Fig 1.11	01

1.12 4	An op-amp has differential gain of $100dB$ and CMRR of $90dB$. If the differential input is $80\mu V$ and the common input is $100mV$, the output voltage= 8.3612 V	01
1.13	In a voltage follower using an op-amp, whose slew rate is $3V/\mu s$, the peak value of the input sinusoidal voltage is $8V$. The maximum frequency of the input signal, so that the output is not distorted= 59.68K	01
1.14 4	An op-amp with finite open loop gain has a unity gain band width of $6MHz$. If it is used as an amplifier with a gain= -59, the $3dB$ frequency of the closed loop amplifier is 100 K .	01
1.15 4	An amplifier has a open loop gain of $40dB$ and an output impedance = $1M\Omega$. If 9.0% current-series negative feedback is provided to this amplifier, the output impedance of the closed loop amplifier= 10 M.	01
1.16 4	An amplifier has an open loop gain of $60dB$ and an input impedance=$1K\Omega$. If 4.9% voltage shunt negative feedback is provided to this amplifier, the input impedance of the closed loop amplifier is 20Ω.	01
1.17 5	A class B complementary symmetry power amplifier operated with a single supply of $6V$ and a loud speaker load of 3Ω. The maximum power output possible= 1.5 W	01
1.18 5	A power transistor with $T_j(max) = 200°C$ has $\theta_{JA} = 5°C/W$. $P_{d(max)}$ for the transistor at an ambient temperature of 50°C= 30 W .	01

PART-B

2	a	Draw the small signal equivalent of the circuit shown in fig 2a and determine the voltage gain $\frac{v_o}{v_i}$, the output impedance, assuming the MOS transistors are operating in saturation.  Fig 2a	06
	b	Design the circuit in fig 2b, so that the NMOS device operates in saturation with $I_D = 2mA$ and $V_D = 3.4V$, and it is $0.6\,V$ away from the edge of triode region.  Fig 2b	10

OR

3 a) In the RC coupled CS amplifier circuit of fig 3a, determine I_D, g_m and the mid frequency gain, $\frac{v_o}{v_s}$, of the amplifier, the lower cut off frequencies f_{L1}, f_{L2}, f_{LS} due to C_{C1}, C_{C2}, C_S respectively, and the upper cut off frequencies f_{Hi} and f_{Ho} due to input and output capacitances respectively. The NMOS device has its $C_{gs} = 2pF$ and $C_{gd} = 0.4pF$.

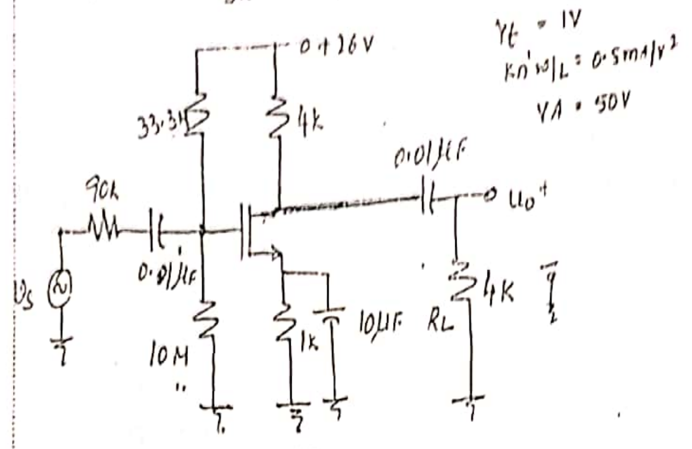


Fig 3a

10

b) In the circuit of fig 3b, calculate the values of V_1, V_2 and V_3. The two transistors are identical with $V_t = 1V, K_n'W/L = 0.25mA/V^2$.

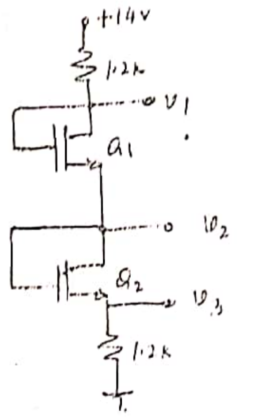


Fig 3b

06

4 Design the values of R_C, R_E, R_1, R_2 in the amplifier circuit of fig 4, so that the operating point is fixed at $V_{CE} = 5V$, $I_C = 2mA$, with bias stabilization factor $S(I_{CO}) = 12$. and the mid frequency small signal voltage gain $\frac{v_o}{v_{sig}} = -120$. Design the values of , C_{C1}, C_{C2} and C_E so that lower cut off frequency of the amplifier is $100Hz$.[Hint: choose C_{C1} and C_{C2} so that the lower cut off frequencies because of C_{C1} and C_{C2} are at least a decade lower than $100Hz$].

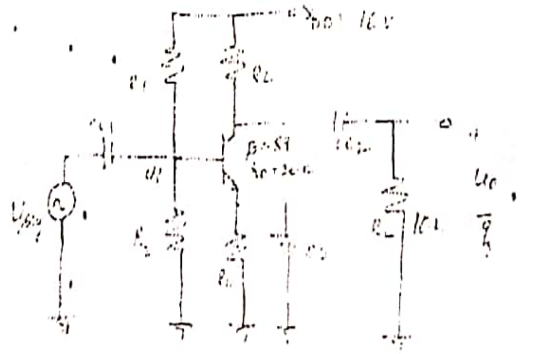
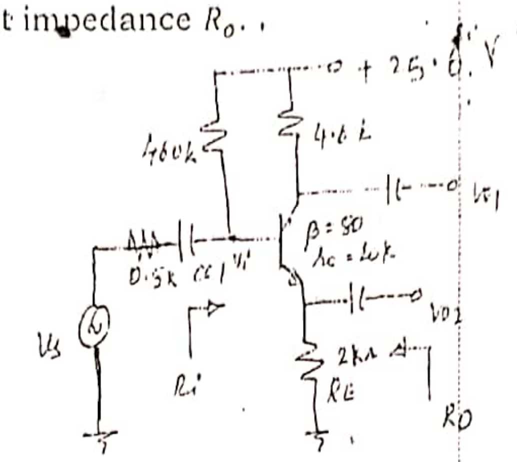
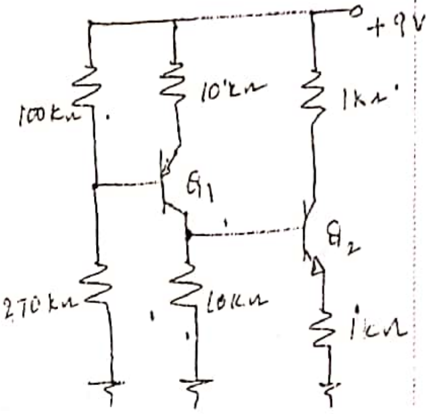
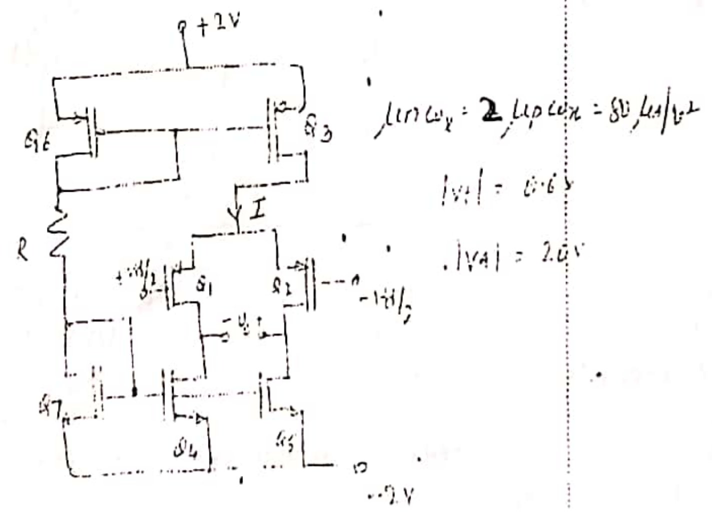


Fig 4

16

OR

5	a	For the circuit of fig 5a, determine the voltage gains $\frac{V_{o1}}{V_s}, \frac{V_{o2}}{V_s}$, input impedance R_i, output impedance R_o.  Fig 5a	08
	b	In the circuit of fig 5b, determine the voltages at all nodes and currents through each of the transistor. Assume $\beta_1 = \beta_2 = 100$.  Fig 5b	08
6	a	In the circuit of fig 6a, Q_1 and Q_2 form a differential pair, while Q_4 and Q_5 form the active loads. Determine the $\frac{W}{L}$ ratios of all the transistors to meet the following specifications: i) $A_d = 60$ ii) $I = 0.2mA$ iii) $V_{G3} = +1.0V$ iv) $V_{G5} = -1.0V$ Neglect channel length modulation for dc calculations. Assume Q_3 and Q_6 are identical.  Fig 6a	10

0 b Design a Widlar current mirror to provide an output current of $20\mu A$. The reference current is $80\mu A$. The BJTs used have very high β and $V_{BE} = 0.7$ at $1mA$ of current. Also find output resistance of the mirror, if $\beta = 100$ and $V_A = 80V$. 06

OR

7 a A current mirror loaded MOS differential amplifier of fig 7a, is biased with current source $I = 1mA$. The NMOS devices have $V_{OV} = 0.3V$ while that of PMOS devices is $0.5V$. $V_{AN} = |V_{AP}| = 10V$. Total capacitance at the input node of the current mirror is 0.08pF and that at the output node of the amplifier is $0.2pF$. Find the value of the differential gain at low frequencies and the frequencies of poles and zero of the differential gain.

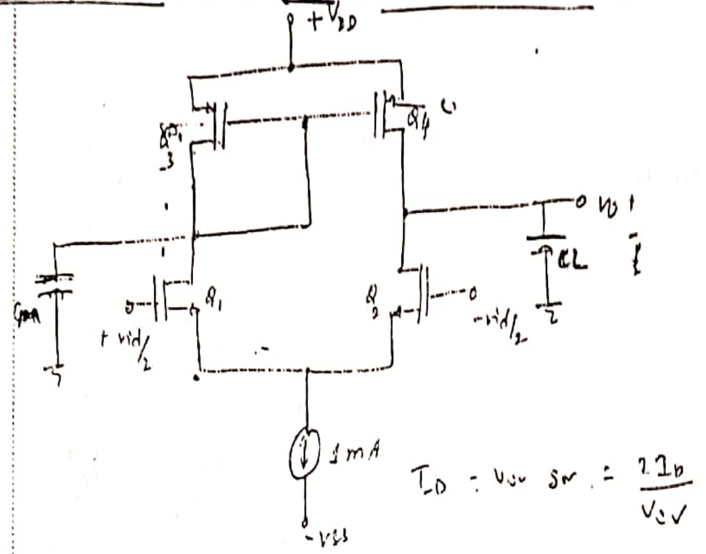


Fig 7a 08

b For the differential amplifier circuit shown in fig 7b, find the differential input impedance R_{id}, differential gain $\frac{v_o}{v_s}$, the common mode input impedance R_{icm}, the common mode gain A_{CM} and CMRR, if the drain resistors are of 1% tolerance. Assume $\beta = 100$ and $V_A = 50V$ for the transistors.

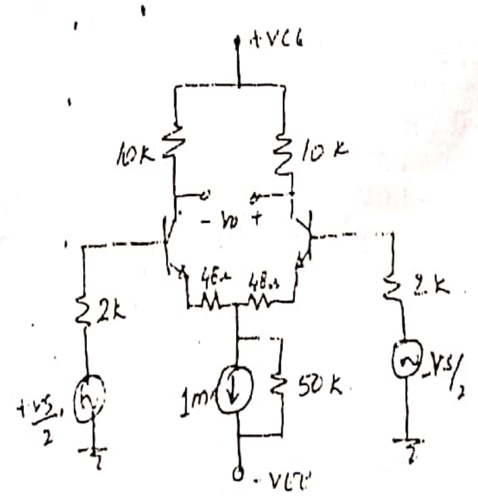
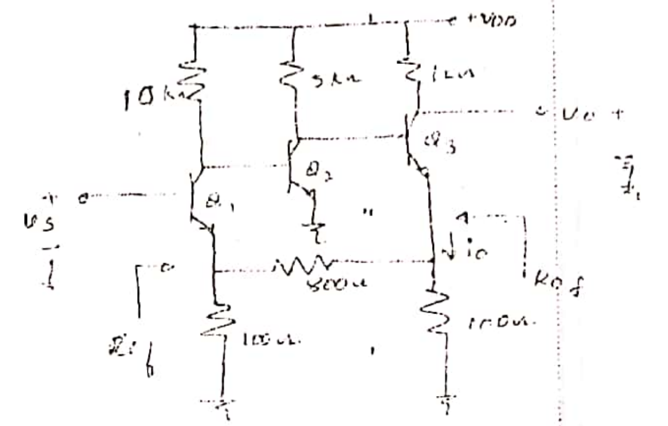
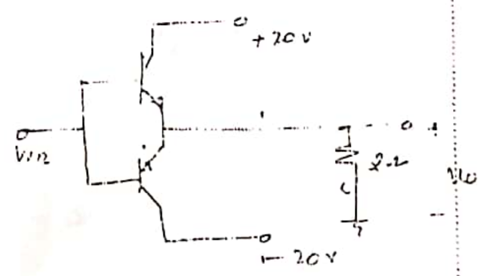


Fig 7b 08

8 a Design a Schmitt trigger circuit using an ideal op-amp, to obtain an $UTP = -2V$ and $LTP = -5V$. Assume that the op-amp is operating with $+12V$ and $-12V$ supplies. 08

b Design a second order active low pass filter with an upper cut off frequency of $3KHz$ and a maximally flat response using Sallen-Key structure. 08

OR

9	a	Design an instrumental amplifier to realize a differential gain, variable in the range 1 to 100 utilizing a 50K pot as a variable resistor.	08
	b 04	i) Design an inverting amplifier with a closed loop gain of 80 and an input impedance of $1K\Omega$ using an ideal op-amp. ii) If the op-amp has an open loop gain of 2000, find the closed loop gain of the amplifier. iii) Design the value of the resistor one can place in parallel with the input resistor to restore the closed loop gain to its original value.	08
10	a	In a complementary symmetry class B power output stage, show that the maximum conversion efficiency is 78.5%.	04
	b 05	In the circuit of fig 10b, if the DC emitter currents of transistors Q_1, Q_2, Q_3 are 1mA, $2mA$ and $4mA$ respectively, find two open loop gain A, the feedback factor β, the closed loop gain $\frac{i_o}{v_s}$, the voltage gain $\frac{v_o}{v_s}$, input impedance R_{if} and output impedance R_{of}. Assume $\beta = 100$ and $r_o = \infty$ for all the transistors.  Fig 10b **OR**	12
11	a 05	A trans conductance amplifier has an input Resistance R_i and output resistance R_o and voltage gain A. If negative voltage series feedback is provided to such an amplifier with feedback factor β, derive the expressions for the input and output impedances of the amplifier with feedback. Assume that there are no loading effects of the feedback network.	05
	b	For the circuit shown in fig 11b, calculate input power, output power and power dissipated by each transistor and the circuit efficiency for an input of $10V$ rms.  Fig 11b	07
	c	A silicon transistor, whose $T_j(max) = 200^\circ C$, can dissipate a maximum power of $100W$ at a case temperature of 140°C. This transistor is connected to a heat sink, whose $\theta_{SA} = 1.5°C/W$, with an insulating washer of $\theta_{CS} = 0.5°C/W$. What maximum power the transistor can dissipate at an ambient temperature of 40°C?	04

SCHEME AND SOLUTION

SUBJECT CODE: 18EC33 SUBJECT: Analog Microelectronics Circuits. [AMC]

Question No	PART - A	Mark
1 1.1	1.2 mA.	01
1.2	4	01
1.3	$I_D = 1mA$, $W/L = 500$	01+0
1.4	$\frac{r_0}{r_e} = \frac{52k}{26} = 2k$	01
1.5	$V_C = 6.7V$, $R_C = 10K\Omega$	01+01
1.6	0.554 V (0.733)	01
1.7	$R_o = g_{m_2} r_{o_2} \left[\frac{1}{g_{m_1}} \| r_{o_1}\right] = 117.073 k\Omega$ (148.047kΩ)	01
1.8	$V_o = 4 \times g_{m_1} \times v_{in} \times 3k$ $V_o/v_{in} = 4 \times 1m \times 3k = 12$	01
1.9	~~$I_0 = 20 mA$~~ $I_0 = 15mA$	01
1.10	$\frac{V_{od}}{v_{id}} = 15$	01
1.11	$V_0 = 6V$	

SCHEME AND SOLUTION

SUBJECT CODE: 12EC33 SUBJECT:

Question No		Marks
1.12	$V_0 = 8.3162$ V	01
1.13	$f_{max} = 59.68$ KHz.	01
1.14	$\omega_{3dB\,(C.L)} = 100$ kHz.	01
1.15	$R_{of} =$ [illegible] $\left[R_{of} = R_o(1+A\beta)\right] = 10\,M\Omega$	01
1.16	$R_{if} =$ [illegible] $\left[R_{if} = R_i / (1+A\beta)\right] = 20\,\Omega$	01
1.17	max. o/p power $= \frac{V_{CC}^2}{8 R_L} = 1.5$ W	01
1.18	$P_D\,max = \frac{T_{J(max)} - T_A}{\theta_{JA}} = 30$ W	01

SCHEME AND SOLUTION

SUBJECT CODE: 12EC43 SUBJECT:

Question No	PART – B	Marks
2 a)	Small signal model	02
	$\frac{v_o}{v_i} = \frac{(g_{m_1} - g_{m_2})(r_{o_1} \parallel r_{o_2})}{1 + g_{m_1}(r_{o_1} \parallel r_{o_2})}$	02
	where $I = v_i[g_{m_2} - g_{m_1}] + g_{m_1} v_o$	01
	$\frac{v_o}{v_i} = 0.399\ V/V.$	01
b)	$I_D = \frac{1}{2} k_n' \, W/L (V_{ov})^2$	01
	$V_{OV} = 4V$	01
	$V_{DS} = V_{GS} - V_t + 0.6$ $V_{DS} = 4.6V$ $\quad V_D = 3.4$	02
	$V_S = -1.2V$	01
	$V_g = 3.8V$	01
	$R_1 = 620k\Omega$	01
	$R_2 = 380k\Omega$	01
	$R_D = \cancel{1.7k\Omega}\ 3.3K$	01
	$R_S = \cancel{1.2k\Omega}$ 600Ω	01

$$x = \frac{0.5}{2}(78 - x - 1)$$

$$4x = \cancel{60.84}\ 46.24 + x^2 - 13.6x$$

$$4x = 25 + x^2 - 10x$$

SCHEME AND SOLUTION

SUBJECT CODE: 12 EC33 SUBJECT:

Question No		Marks
3 a)	$I_D = \frac{1}{2} \mu_n C_{ox}' W/L\ V_{ov}^2$	
	$\boxed{I_D = 2mA}$ 2.1 mA and $\boxed{V_g = \frac{26 \times 10M}{43.3M} = 6V}$	01
	$g_m = \frac{2I_D}{V_{ov}} = 2mA/V = 1.45 mA/V$. $V_{ov} = 6 - 2.1 - 1$	
	$r_0 = 25K = 23.8 K\Omega$.	
	$\frac{V_0}{V_i} = -g_m [4K \Vert 4K \Vert 25K] = -3.7 = -2.675$	01
	$\frac{V_i}{V_S} = \frac{7.7M}{90K + 7.7M} = 0.987 \quad 0.988$	01
	$\frac{V_0}{V_S} = \frac{V_0}{V_i} \times \frac{V_i}{V_S} = -3.652\ V/V = -2.6429$	01
	$f_{LCC_1} = \frac{1}{2\pi C_{C1}[R_S + R_G]} = 2.043 Hz$	01
	$f_{LCC_2} = \frac{1}{2\pi C_{C_2}[R_D \Vert r_0 + R_L]} = 2.143 KHz$ $2.136 KHz$	01
	$f_{LCS} = \frac{1}{2\pi C_S[\frac{1}{g_m} \Vert R_S]} = 32 Hz$ $38.99 Hz$	01
	$f_{Hi} = \frac{1}{2\pi C_{in}[R_S \Vert R_G]} = 463.48 KHz$ $515.57 KHz$.	01
	$C_{in} = 3.86 PF$. 3.47	
	$f_{Ho} = \frac{1}{2\pi C_o[R_D \Vert r_0 \Vert R_L]} = 214.859 MHz$ $157.1 MHz$	01
	$C_o = 0.549 pf$	

SCHEME AND SOLUTION

SUBJECT CODE: 12EC33 SUBJECT:

Question No		Marks
b)	$I_D = \frac{0.25\times10^{-3}}{2}(V_2 - V_3 - V_t)^2 \rightarrow (1)$	01
	$I_D = \frac{0.25\times10^{-3}}{2}(V_1 - V_2 - 1)^2 \rightarrow (2)$	01
	$V_1 + V_3 = 2V_2$	01
	$V_2 = 7V.$	01
	$V_1 = 11.816 V$	01
	$V_3 = 2.184 V$	01
4	$A_V = \frac{-R_C \parallel R_L \parallel r_o}{r_e}$	01
	$r_e = \frac{26}{2.02m} = 12.855 \Omega$	01
	$R_C = 2k\Omega$	01
	$R_E = 3.465 k\Omega$	01
	$S(I_{CO}) = \frac{(1+\beta)\left(1+\frac{R_{th}}{R_E}\right)}{1+\beta+R_{th}/R_E}$	01
	$R_{th} = 44.13 k\Omega$	01
	$V_{th} = 8.681 V$	02
	$R_1 = 81.33 k\Omega$	01
	$R_2 = 96.478 k\Omega$	0

Question No		Marks
	R a/c $C_E = (r_e \| R_E)$ $(R_S = 0)$ $\cancel{R_E \| (r_e + R_B/\beta)}$	01
	$R = 12.8\Omega$ $\cancel{13.29}$	
	$\boxed{C_E = \frac{1}{2\pi \times 100 \times 12.8} = 124.34\mu F}$	01
	C_{C1} and C_{C2} are 10 Hz.	
	R a/c $C_{C1} = R_B \| (\beta+1) r_e = 1.122K$	01
	$\boxed{C_{C1} = 13\mu F}$	01
	R a/c $C_{C2} = R_C \| r_o + R_L = 11.818K$	01
	$\boxed{C_{C2} = 1.347\mu F}$	01
5 a)	$I_B = \frac{25.6 - 0.7}{460K + (\beta+1)R_E} = 40\mu A$ $\cancel{38.9\mu A}$ $40.1\mu A$	01
	$I_E = 3.24mA$, $r_e = 8.024\Omega$	01
	$\frac{V_{o1}}{V_i} = A_V = \frac{-\beta' R_C}{(\beta+1) r_e + (\beta'+1) R_E}$ where $\beta' = \frac{\beta r_o}{R_C + R_E + r_o} = 60$	01
	$\frac{V_o}{V_i} = -2.25$	01
	$\boxed{A_{V_1} = -2.287}$	
	$\frac{V_{o2}}{V_i} = A_{V_2} = \frac{R_E'}{R_E' + r_e}$ where $R_E' = R_E \| r_o$	

Question No		Marks

$R_E' = 1.818\,k\Omega$

$\boxed{A_{V_2} = \frac{v_{o2}}{v_i} = 0.995}$ 01

$R_i = (\beta+1)[r_e + R_E'] \,||\, R_B$

$R_i = 147.9k \,||\, 460k = 111.922\,k\Omega$ 01

$R_o = \frac{R_S \,||\, R_B}{\beta+1} + r_e \,||\, R_E = 14.09\,\Omega$ 01

$\frac{v_{o1}}{v_s} = \frac{v_{o1}}{v_i} \times \frac{v_i}{v_s} = -2.287 \times 0.995 = -2.275$

$\frac{v_{o2}}{v_s} = \frac{v_{o2}}{v_i} \times \frac{v_i}{v_s} = 0.990$ 01

5) For Q_1 02

$V_{th} = 6.567\,V$ and $R_{th} = 73\,k$ $\quad V_{E_1} =$

$\boxed{I_{B_1} = 1.6\,\mu A}$, $\boxed{I_{C_1} = 0.16\,mA}$, $\boxed{I_{E_1} = 0.1616\,mA}$ 02

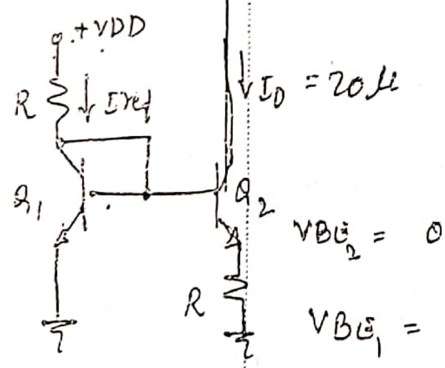
$1.73\,\mu A$ $\quad 0.1733\,mA$ $\quad 1.75\times10^{-4}\,A$

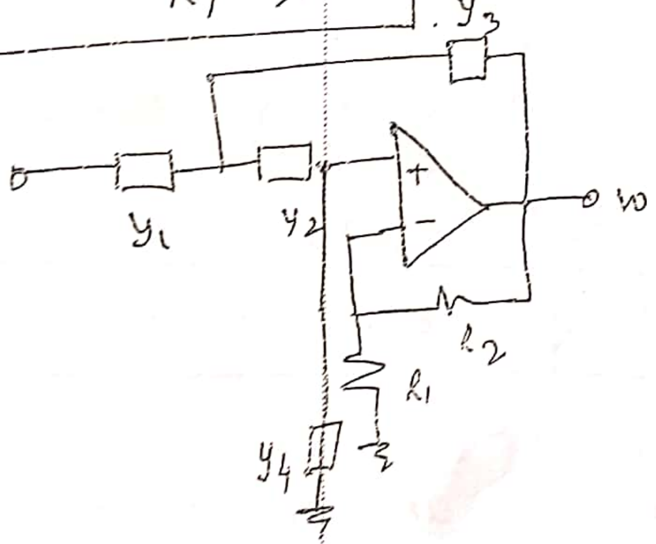
$V_{C_1} = V_{B_2} = 1.5\,V$ $\quad V_{CE_1} = -5.768\,V$ $\quad V_{E_1} = 7.267\,V$ 01

For Q_2

$V_{E_2} = 0.809$, $\quad V_{CE_2} = 7.177\,V$ 01

$I_{E_2} = 0.809\,mA$, $\quad I_{B_2} = 8.00\,\mu A$, $\quad I_{C_2} = 0.801\,mA$ 02

Question No		Marks
6 a)	$R = \frac{V_{G6} - V_{G5}}{I} = 10\,k\Omega$	01
	$I_{D3} = \frac{1}{2} k_p' (W/L)_3 V_{ov}^2$ $(W/L)_3 = (W/L)_6 = 62.5$	02
	$I_{D5} = \frac{1}{2} k_n' (W/L)_5 V_{ov}^2$ $(W/L)_5 = \cancel{(W/L)_7} = (W/L)_4 = 15.625$	02
	$(W/L)_7 = 31.25$	
	$A_d = g_m R_d = g_m [r_{o1} \parallel r_{o4} = r_{o2} \parallel r_{o5}] = 100K \times g_m$	02
	$g_m = 2 I_D / V_{ov}$ $(V_{ov})_{1,2} = -0.333$ $\boxed{g_m = 0.6\,mA/V}$	01
	$I_{D1} = \frac{1}{2} k_p' (W/L)_1 V_{ov}^2$ $(W/L)_1 = (W/L)_2 = 45.09$	02
b)		
	Ckt	01
	$V_{BE_2} = 0.598\,V$ $V_{BE_1} = 0.634\,V$	01
	$R = \frac{V_E}{20\mu} = \frac{0.634 - 0.598}{20\mu} = 1.8K$	01
	$R_o = R_E' + (1 + g_m R_E') r_o$ $r_o = 4M$ and $r_\pi = 1.3\,k\Omega$	01
	$R_E' = (\beta+1) r_e \parallel R_E = 1.775\,k\Omega$	01
	[illegible]	[illegible]

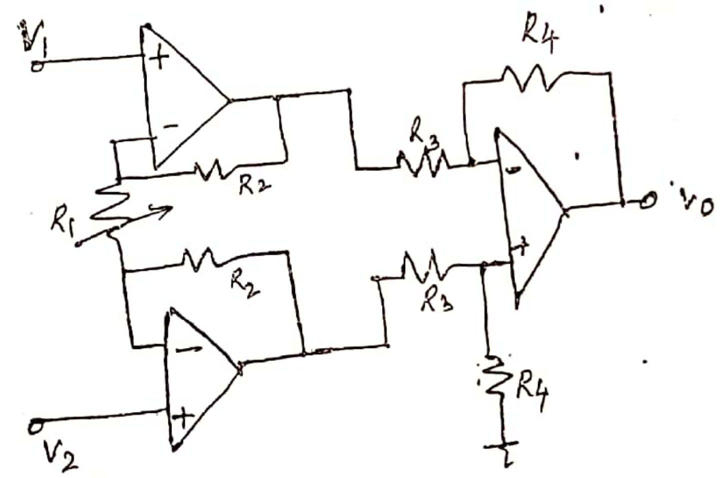
Question No		Marks
8 a)	UTP = −2 V LTP = −5 V	
	$UTP = V_R\left(\frac{R_1}{R_1+R_2}\right) + V_{CC}\frac{R_2}{R_1+R_2}$ or $V_R + (V_{CC} - V_R)\beta$	01
	$LTP = V_R\frac{R_1}{R_1+R_2} - V_{CC}\frac{R_2}{R_1+R_2}$ or $V_R - (V_{CC} + V_R)\beta$	01
		03
	$V_R = -3.88$ V. and $\beta = \frac{R_2}{R_1+R_2}$. $= -4$ Design	
		01
	$\beta = 0.125$	01
	$R_2 = 1K$	01
	$R_1 = \cancel{9K}$ 7K	
b)		02
	$\frac{V_o}{V_i} = \frac{A(Y_1Y_2)}{Y_1Y_2 + Y_4(Y + Y_2 + Y_4) + Y_3Y_2(1 - A_o)}$	01
	where $A_o = 1 + R_2/R_1$	01

SCHEME AND SOLUTION

SUBJECT CODE: 12 EC33 SUBJECT:

Question No		Ma
7 a)	$g_{m1} = g_{m2} = \frac{2I/2}{V_{OV(N)}} = 3.333\,mA/V$	01
	$g_{m3} = g_{m4} = \frac{2I/2}{V_{OVP}} = 2\,mA/V$	01
	$r_o = 20k$	01
	$A_d = g_{m1}(r_{o2} \parallel r_{o4})$ $= 33.3\ V/V$	02
	$f_{P1} = \frac{1}{2\pi C_L (r_{o2} \parallel r_{o4})} = 198.94\,MHz$ 79.57 MHz	01
	$f_{P2} = \frac{1}{2\pi \frac{C_m}{g_{m3}}} = 1.591\,GHz$ 3.97 G	01
	$f_Z = 2f_{P2} = 3.183\,GHz$ 7.95 G.	01
b)	$R_{id} = 2r_e(\beta+1) = 20.2\,k\Omega$ and $r_e = 52+48 = 100\Omega$	02
	$\frac{V_o}{V_S} = \frac{V_o}{V_{id}} \times \frac{V_{id}}{V_S} = g_m(R_C \parallel r_o) \times 0.8347$ $\boxed{\frac{V_o}{V_S} = 90.90 \times 0.8347 = 75.88}$	02
	$R_{icm} = (\beta+1)\left[R_{EE} \parallel \frac{r_o}{2}\right]$ and $r_o = 100k\Omega$ $\boxed{R_{icm} = 2.525\,M\Omega}$	01
	$\boxed{A_{cm} = \frac{\Delta R_C}{2R_{EE}} = 2m}$	01
	$CMRR = \frac{A_d}{A_{cm}} = 37.94k = \boxed{91.58\,dB}$	02

SUBJECT CODE: 12EC33 SUBJECT:

Question No		Marks
	For Lowpass filter	
	$y_1 = y_2 = \frac{1}{R}$ & $y_3 = y_4 = j\omega C$	01
	$\frac{v_o}{v_i} = \frac{A_0}{\frac{s^2}{\omega_n} + \alpha(s/\omega_n) + 1}$	01
	for maximal flat response $\alpha = 1.414$	
	$RC = \frac{1}{2\pi 3k} = 5.305\times10^{-5}$ $R_1 = 1K$ $R_2 = 0.5K$	01
	If $C = 1\mu F$, $R = 53.05\,\Omega$	01
9a)		01
	$A_d = \frac{R_4}{R_3}\left(1 + \frac{2R_2}{R_1}\right)$	01
	POT reads 50K $1 = \frac{R_4}{R_3}\left(1 + \frac{2R_2}{R_{1f} + 50K}\right)$	01
	$R_4 = 1K$, $R_3 = 2K$	
	POT reads 0Ω $100 = \frac{R_4}{R_3}\left(1 + \frac{2R_2}{R_{1f}}\right)$	
	$2R_2 = R_{1f} + 50K$	
	$199 R_{1f} = 2R_2$	01
	$R_{1f} = 252.5\Omega$ $R_2 = 25.12K$	

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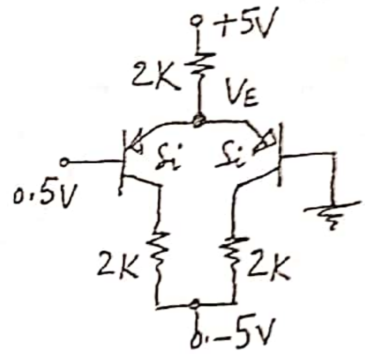
R. V. COLLEGE OF ENGINEERING
Autonomous Institution affiliated to VTU
III Semester B. E. Examinations Nov/Dec-14
Electronics and Communication Engineering
ANALOG MICROELECTRONIC CIRCUITS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	Draw the physical structure (cross-section) of an enhancement type *NMOS* transistor.	02
	1.2	Sketch the frequency response of an $R-C$ coupled common source amplifier and mention the reasons for the gain fall at signal frequencies below and above the mid band.	02
	1.3	Draw the r_e model for a *BJT* operating in *CE* configuration and define the parameters there in.	02
	1.4	Mention two advantages of Widlar current mirror.	02
	1.5	Find V_E in the circuit of fig1.5. Assume $\|V_{BE}\| = 0.7V$.  Fig 1.5	02
	1.6	In a Wilson current mirror using identical *MOSFETs*, the output impedance is ____________.	02
U4	1.7	Define slew rate and unity gain *BW* of an opamp	02
U4	1.8	Draw the circuit of a super diode using an ideal opamp.	02
	1.9	Write the Barkhausen criteria for sustained oscillations.	02
	1.10	Draw the collector current wave forms for a transistor operating in: i) Class *A* ii) Class *B*.	02

PART-B

2	a	Write the expressions for the drain current of an enhancement *NMOS* transistor in both ohmic and saturation regions. From the above obtain the expressions for r_{DS} and g_m.	08

$r_{DS} = \frac{1}{k_n' W/L (V_{GS} - V_t)}$

$g_m = k_n' W/L (V_{GS} - V_t)$

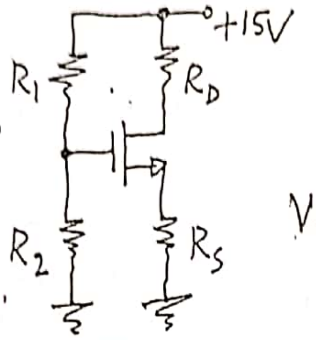
b) Design the circuit of figure 2b to establish a dc drain current $I_D = 0.5mA, V_S = 4V$ and $V_{DS} = 5V$. The $MOSFET$ is specified to have $V_t = 1V$ and $K_n' \frac{W}{L} = 1mA/V^2$. Calculate the percentage change in the value of I_D obtained when the $MOSFET$ is replaced with another having the same $K_n' \frac{W}{L}$ but $V_t = 1.5V$. Assume $V_{DD} = 15V$ and $\lambda = 0$.

$R_D = 12k$

$R_S = 8k$

$R_2 = 6M$

$R_1 = 9M$



$V_G = 6V.$

Fig 2b 08

$\Delta I_D = 0.44mA$

% change = 12%

OR

3 a) Draw a simple high frequency model for a $MOSFET$ and obtain an expression for its unity gain frequency, f_t. 08

b) In the circuit of fig 3b, determine V_1, V_2 and V_3. Assume Q_1 and Q_2 are identical.

$V_1 = 4.8V$

$V_2 = 2V$

$V_3 = 0.8$

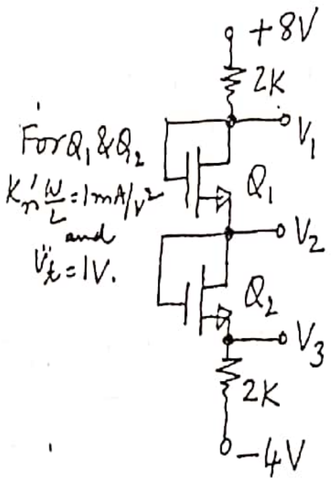


Fig 3b 08

4 a) For the circuit shown in figure 4a, $V_{CE} = 3.2V$ when $V_B = 0.7V$. Determine the new value of V_B' required so that the transistor operates at the edge of saturation. Assume $V_{CE}(Sat) = 0.3V$.

$V_{BE} = V_B = 0.7043$

$I_{C1} = 1mA$

$I_{C2} = 1.43mA$

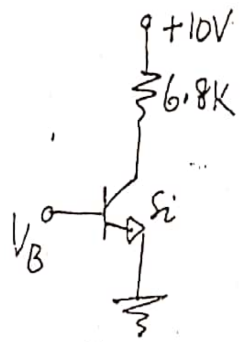


Fig 4a 04

b) With a suitable circuit diagram, explain how a Bipolar junction transistor could be used as a switch. 04

c For the circuit shown in figure 4c, determine the voltages at all and currents through all branches. Assume all are *Si* transistor: very large β. **08**

$V_{C_1} = 7.8V$
$V_{E_1} = 4.3V$
$V_{B_1} = 5V$
$I_{E_1} = 1.4$

$V_{B_2} = 7.8$
$I_{E_2} = 3.4m$
$V_{C_2} = 4.8$, $V_{E_2} = 8.5$.

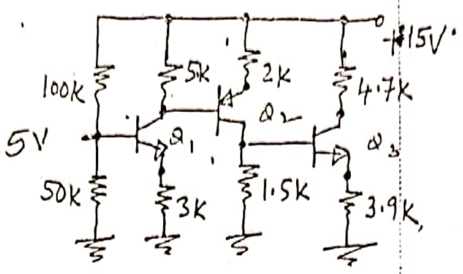


Fig 4c

Q_3
$I_{E_3} = 1.06mA$
$V_{E_3} = 4.15V$
$V_{C_3} = 10V$

OR

5 a Determine the values of R_C, R_E, R_1 and R_2 in the circuit of Fig 5a, so that $V_{CE} = 6V$, $I_E = 1mA$, $S(I_{CO}) = 10$ and $\frac{v_o}{v_i} = -100$. Assume $\beta = 99$ for the transistor. **10**

$R_C = 2.6k$
$R_E = 3.426k$
$R_{th} = 34.26k$
$V_{th} = 4.446V$
$R_1 = 92k$,
$R_2 = 54.55k$.

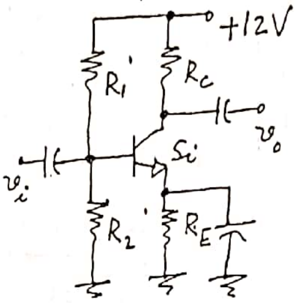


Fig 5a

$S_{ICO} = \frac{(\beta+1)(R_B+R_E)}{R_B+(\beta+1)R_E}$

b Draw the high frequency small signal equivalent circuit for a *BJT* and explain the significance of each component. **06**

6 a In the active loaded *MOS* differential amplifier circuit shown in fig 6a, the *NMOS* devices are identical with $K_n' \frac{W}{L} = 1mA/V^2$ and $V_{AN} = 30V$ and the *PMOS* devices are identical with $K_p' \frac{W}{L} = 0.64\ mA/V^2$ and $V_{AP} = 40V$, determine: **10**

i) $g_{m1}, g_{m2}, g_{m3}, g_{m4}, r_{o1}, r_{o2}, r_{o3}$, and r_{o4}
ii) Differential gain.
iii) Common mode gain and *CMRR*.
iv) The upper cut-off frequencies of the differential gain. The effective values of C_m and C_L are indicated in the figure.

$g_{m_1} = g_{m_2} = 1mV$
$g_{m_3} = g_{m_4} = 0.8mV$
$r_1 = r_2 = 60k$
$r_3 = r_4 = 50k$.
$A_d = 34.28$
$A_{cm} = -6.25m$

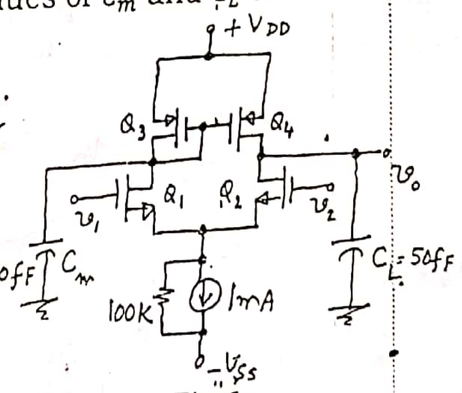


Fig 6a

$f_{p_1} = 92.8M$ $f_{p_3} = 2.29Hz$.
$f_{p_2} = 2.12G$ $CMRR = 74.784dB$

	b	Determine the values of R and R_E in the Widlar mirror circuit of fig 6b assuming β is large. Also determine the output impedance of the current source, if $\beta = 100$ and $V_A = 25V$.	06

$I_C = 1mA$ @ 0.7 at Room temp. 1.8V

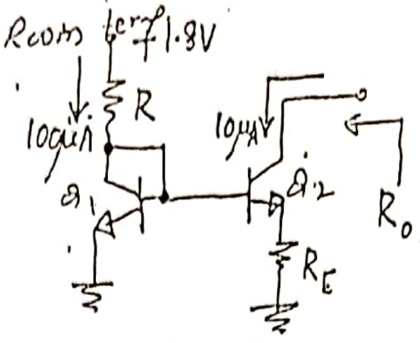
$V_{BE_1} = 0.64$.

$V_{BE_2} = 0.58$

$R = 11.6K$.

$R_E = 6K$.

$r_e = 2.6K$ | $R_E' = 5.864K$ OR, $R_o = 2.5M$.



$R_o = R_E' + (1 + g_m R_E')$

$R_o = 8.14 M\Omega$

Fig 6b

7	a	For the differential amplifier circuit shown in figure 7a, determine the following, assuming β for all transistors is 99.	10

i) The input differential resistance R_{id}
ii) The differential voltage gain.
iii) The worst case common mode gain, assuming $\Delta R_C = 0.02\, R_C$.
iv) The CMRR in dB and
v) The input common mode resistance, assuming an Early voltage $V_A = 100V$.

$r_e = 100\,\Omega$

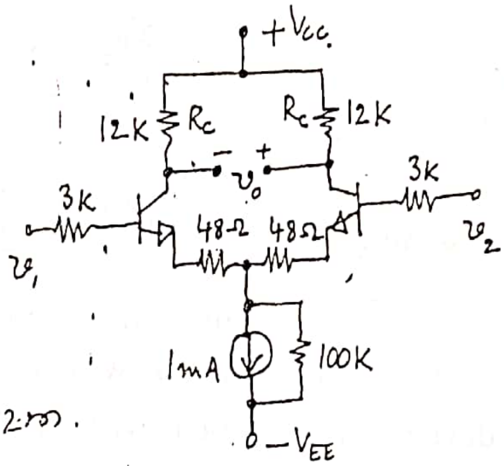
$R_{id} = 20K$

$\frac{V_o}{V_i} = 120$,

$A_d = \frac{V_o}{V_s} = \frac{20 \times 120}{26} = 92.3$

$A_{cm} = \frac{V_o}{V_{icm}} = \frac{\Delta R_C}{2R_{EE}} = 1.2m$.

CMRR = 97.7dB



$R_{icm} = 5M\Omega$

$r_o = 200K$

Fig 7a

	b	Draw the circuit of a Wilson current mirror using $BJTs$ and obtain the expression for the output impedance.	06
8	a	The input v_i to the circuit shown in fig8a is a low frequency sine wave signal of peak voltage V_p. The opamp is specified to have output saturation voltages of $\pm 13V$ and output current limits of $\pm 20mA$.	

i) $V_o = -6V$.

ii) $V_{th} = 2.167$.

iii) $V_o = 13V$

iv) $R_L = 620\,\Omega$

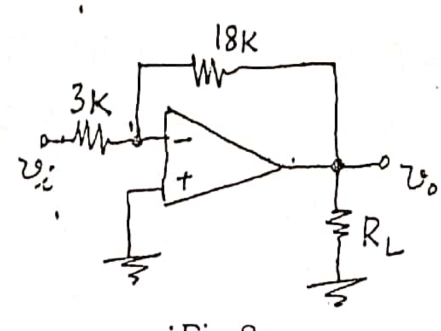


Fig 8a

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Nov/Dec-15

Electronics and Communication Engineering

ANALOG MICROELECTRONIC CIRCUITS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	The region of operation of a P-channel MOSFET, with $V_t = -1V$, is Sat, when it is biased such that $V_G = -5V$, $V_S = -1V$ and $V_D = -6V$.	01
	1.2	An NMOS transistor acts as an 500 Ω resistor for small values of V_{DS}, when $V_{OV} = 2V$. If the device has to act as a 400 Ω resistor, V_{OV} 2.5.	01
	1.3	In the circuit of fig 1c, the minimum value of V_{DD} so that Q_1 does not enter triode region is equal to 1.5.	01

10%6::100

500:2::400:?

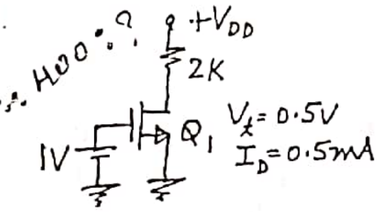


fig 1c

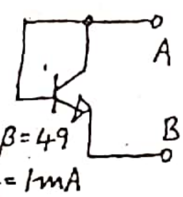


fig 1g

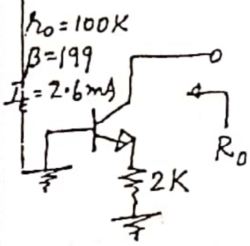


fig 1h

1	1.4	An NMOS transistor, whose $C_{OX} = 10\ \frac{fF}{\mu^2}$, $W = 50\mu$, $L = 0.4\mu$ and $V_t = 1V$, is biased with V_{GS} equal to $2V$. The total charge stored in the channel for $V_{DS} = 0$ is 200fC	01
2	1.5	The intrinsic gain of a BJT, whose early voltage $= 26V$, at $27°C$ = -1000.	01
2	1.6	A BJT operated in CC configuration acts closer to a voltage amplifier.	01
2	1.7	Assuming the transistor to be operating in the active region, the incremental resistance between the terminals A and B in the circuit of fig 1g = 26Ω.	01
2	1.8	Assuming the transistor to be operating in the active region, the output impedance, R_o in the circuit of fig 1h = 10MΩ	01
2	1.9	In the mirror circuit of fig 1i, the transistors are identical with $\beta = 40$. I_O = 95.24μ	01
3	1.10	In the differential amplifier circuit of fig 1j, the transistors are identical with $V_t = 0.7\ V$ and $k_n'\ W/L = 3.5\ mA/V^2$. If $V_{CS}(min) = 0.5V$, $V_{ICM}(min)$ and $V_{ICM}(max)$ for the differential amplifier are respectively -0.8528 and 2.5.	02

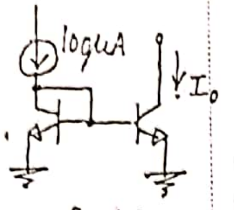


fig 1i

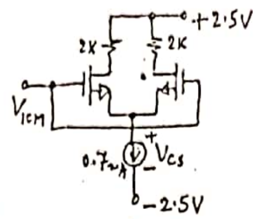


fig 1j

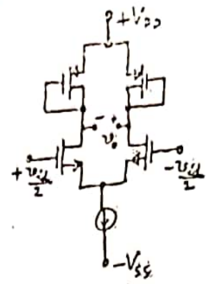


fig 1k

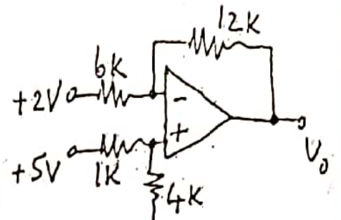


fig 1l

2 1.11 In the circuit of fig1k, v_o/v_{id} = <u>2</u>. Assume $g_{mn} = 2\,mA/V$, $g_{mp} = 1\,mA/V$ and $\lambda = 0$, for all transistors. 01

3 1.12 The output voltage in the circuit of fig1l = <u>8 V</u>. 01

3 1.13 A difference amplifier has a differential gain of $100\,dB$ and $CMRR = 88\,dB$. If the output voltage is $200\,mV$ and the common mode input is $25\,mV$, the differential input = <u>1 μV</u>. 01

4 1.14 In a voltage amplifier using an op amp, whose unity gain bandwidth is $5\,MHz$ and slew rate is $10\,V/\mu sec$, the maximum amplitude of a step input, so that the output is purely exponential, is equal to <u>0.318 V</u>. 01

4 1.15 An op amp with a unity gain bandwidth of $10\,MHz$ is used as an amplifier with a gain of 20. The $3\,dB$ frequency of the closed loop amplifier = <u>500 KHz</u> 01

4 1.16 An amplifier delivers a power output of $10\,W$ with 15% total harmonic distortion, when the input voltage is $15\,mV$. If $40\,dB$ negative feedback is provided and the output power is to remain at $10W$, the required input signal = <u>1.5 V</u> and the total harmonic distortion of the closed loop amplifier = <u>0.15%</u>. 02

$T_j - T_A = \theta_{JA} P_D$

$T_j = T_A + \theta_{JA} \cdot P_D$

5 1.17 A power transistor has its $\theta_{JA} = 2^0\,C/W$. If the dissipation in the transistor is $40\,W$ at an ambiemnt temp of 40°C, the junction temperature of the transistor = <u>120°C</u>. 01

5 1.18 A transformer coupled class-A power amplifier, operating from a $12\,V$ battery, is delivering power to a loud speaker of $4\,\Omega$. If the transformer turns ratio is 2:1, the maximum power output is equal to = <u>4.5 W</u>. 01

$R_L' = \left(\frac{N_1}{N_2}\right)^2 R_L = \left(\frac{2}{1}\right)^2 4 = 16$

PART-B

$P_{ac} = \frac{V_{CC}^2}{2R_L'} = \frac{12^2}{2\times16} = 4.5$ watts

2 (a) In the circuit of fig 2a, given $V_t = 0.5\,V$ and $k_n'\ W/L = 0.5\,mA/V^2$, determine $V_G, V_D, V_S, I_1, I_2, I_D$ and I_S. Assume $\lambda = 0$.

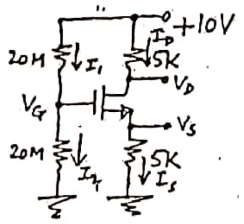


fig2a

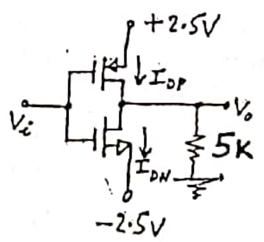


fig2b 08

b Assuming the two transistors are matched with $k_n'\ W/L = k_p'\ W/L = 0.5\,mA/V^2$, $|V_t| = 0.5\,V$ and $\lambda = 0$ in the circuit of fig2b, find the drain currents I_{DP} and I_{DN} and V_o for:

i) $V_i = 0$

ii) $V_i = +2.5\,V$

08

OR

3 a. Calculate the values of V_1, V_2, V_3 in the circuit of fig3a. The $MOSFETs$ are identical with $V_t = 1V$ and $k'_n\ W/L = 0.25\ mA/V^2$.

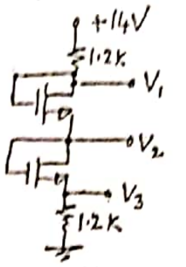


fig3a

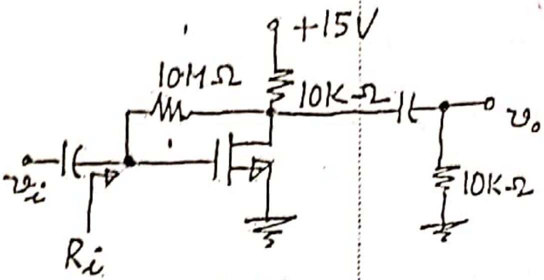


fig3b

08

b. In the CS amplifier circuit of fig3b, determine:

i) v_o/v_i

ii) Input impedance, R_i

iii) The maximum possible peak amplitude of the input signal, so that the output is not distorted. The $MOSFET$ has $V_t = 1.5\ V, V_A = 50\ V$ and $k'_n = 0.25\ mA/V^2$.

08

4 a. In the circuit of fig 4a,

i) Determine the operating point and also the stabilization factor, $S(I_{CO})$.

ii) Determine the voltage gain, v_o/v_s , and the input and output impedances of the amplifier.

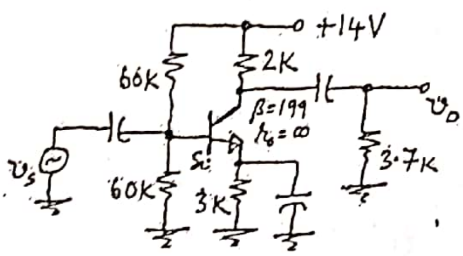


fig4a

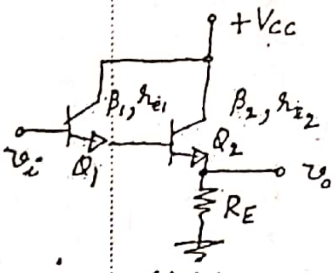


fig4b

08

b. Draw the small signal equivalent of the circuit shown in fig4b and obtain the expressions for the voltage gain, v_o/v_i, the input and output impedances.

08

OR

5 a. Determine I_E in the circuit of fig5a and draw its small signal equivalent and determine $\frac{v_o}{v_s}$,R_{in} and R_0.

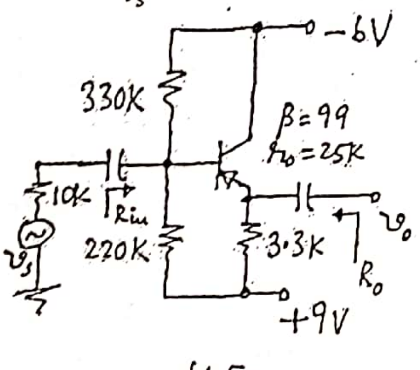


fig5a

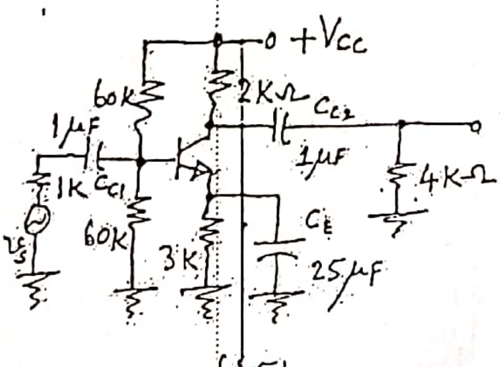


fig5b

08

	b	In the RC coupled amplifier circuit of fig5b, determine the lower cut-off frequencies f_{LC1}, f_{LC2} and f_{LCE} due to C_{C1}, C_{C2} and C_E respectively and the upper cut-off frequencies f_{hi} and f_{ho}, due to input and output capacitances respectively. Also, calculate the bandwidth of the amplifier. The transistor is biased at $I_E = 1\,mA$ and has $\beta = 199$, $r_0 = \infty, C_{be} = 8pF$ and $C_{bc} = 2pF$.	08
6	a	In the Wilson mirror circuit of fig6a, determine I_o, V_{G2}, V_{G3} and the lowest possible V_{o}. Also find g_m and r_0 of Q_2 and Q_3 and the output impedance R_{o}. Given $V_t = 0.6\,V$, $k_n' = 250\,\mu A/V^2$, $3(W/L)_1 = 1.5(W/L)_2 = (W/L)_3 = 60\mu$ and $V_A = 20\,V$ for all the transistors.	08
	b	In the differential amplifier circuit of fig6b, find V_{OV}, g_m of the transistors, the differential gain, the common mode gain and the $CMRR$. Assume the drain resistances are of 0.5% tolerance, $\lambda = 0$ and $k_p' = 1m\,A/V^2$.	08

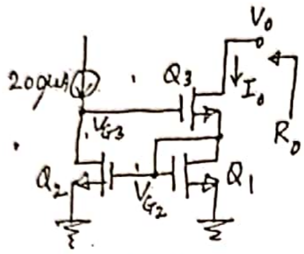


fig6a

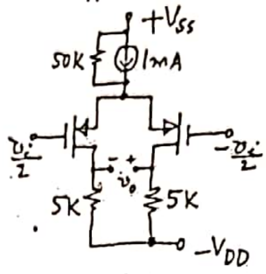


fig6b

OR

7	a	Design a Widlar current mirror circuit for obtaining an output current of $20\mu A$ assuming $V_{BE} = 0.7\,V$ @ $1\,mA$ and the transistors have very high β. The reference current is $0.5\,mA$. If the transistors have $\beta = 100$, $V_A = 20\,V$, determine the output impedance of the circuit.	08
	b	A differential amplifier circuit is shown in fig 7b. Determine the differential input impedance, R_{id}, differential gain, A_d, common mode input impedance, R_{icm}, worst case common mode gain, A_{cm}, and $CMRR$, if the drain resistors are of 0.5% tolerance.	08

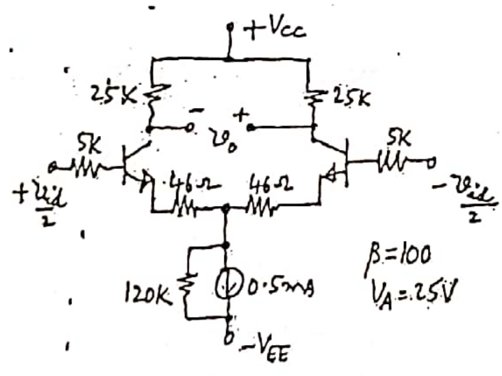


fig7b

8	a	Design a difference amplifier using an ideal op amp to have a differential input impedance of $10\,K\Omega$, differential gain of $28\,dB$ and common mode gain equal to zero. In the designed circuit, if the resistors used are of 0.25 % tolerance, calculate the worst case differential gain, common mode gain and $CMRR$, Also calculate the common mode input resistance, R_{icm}.	10

b Draw the circuit of a full wave precision rectifier using two ideal op amps and explain its operation. The circuit should ensure that none of the op amps go into saturation at any time. 06

OR

9 a Obtain an expression for the voltage gain of the amplifier circuit, using an ideal op amp, shown in fig 9a. Design the values of the different resistors to have an input impedance of $0.5\ M\Omega$ and a gain of -50. Note the maximum value allowed for any resistor is $1\ M\Omega$.

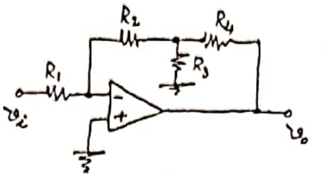


fig 9a

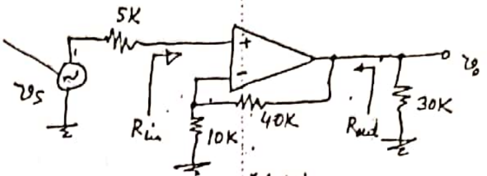


fig 10b

08

b Design an instrumentation amplifier using three ideal op amps to have a differential gain variable from 0.5 to 30 using a $47\ K\Omega$ potentiometer and with zero common mode gain. 08

10 a A current amplifier has a gain A,, input impedance R_i and output impedance R_0. Derive the expressions for the input and output impedances, R_{if} and R_{of}, when current shunt negative feedback is provided to the amplifier with a feedback factor β . 06

b The op amp used in the circuit of fig10b has an open loop gain of 1000, input impedance of $100\ K\Omega$ and output impedance of $10\ K\Omega$. Determine $\frac{v_o}{v_s}$,R_{in} and R_{out} of the closed loop amplifier using feedback theory. 06

c Prove that the conversion efficiency of a class-B power output stage is 50%, when power dissipation in the transistors is maximum. 04

OR

11 a It is required to design an amplifier to have a nominal closed loop gain of 20 using a battery operated amplifier whose gain reduces to half its normal full battery value over the life of the battery. If only 1% drop in closed loop gain is desired, determine the nominal open loop amplifier gain required and the feedback factor. 04

b In a complementary symmetry class-B power output stage, operating with a single supply of $20V$:

i) What is the maximum power output possible into a 8Ω loud speaker?

ii) If the input is $8\,Sin\,\omega t$, what is the power output delivered to the loud speaker?

iii) What is the power dissipation in each of the two transistors, for the input as in (ii)?

iv) What should be the minimum power dissipation capability of each transistor used in such an output stage? 08

c A power transistor for which $T_j(max) = 180^0C$ can dissipate $50W$ at a case temp 50°C. If it is connected to a heat sink ($\theta_{CS} = 0.6^0\ C/W$), what heat sink temp is required to ensure safe operation at $30\ W$ dissipation. For an ambient temp of 39°C, what should be the thermal resistance of the heat sink? 04

SCHEME AND SOLUTION

SUBJECT CODE: 12EC33 SUBJECT: Analog Microelectronic circuits

Question No	PART - A	Marks
1)	a) Saturation b) 2.5V c) 1.5V d) 200fC e) -1000 f) Common Collector g) 26 Ω h) 10 MΩ i) 95.24 μA j) -0.8528 V and 2.5V k) 2 l) 8V m) 1μV n) 0.318 V o) 0.5MHz p) 1.5V and 0.15% q) 120°C r) : 4.5 W Q 1.j and 1.P carry 2 marks. Rest of them carry 1 mark each	20M

PART - B

2a)	$V_G = \frac{10 \times 20M}{20M + 20M} = 5V$ (1M) ; $\therefore I_D = \frac{K_n'(\frac{W}{L})}{2}(V_{GS} - V_t)^2$ (1M) $I_D(mA) = 0.25(4.5 - 5I_D)^2$; $25I_D^2 - 49I_D + 20.25 = 0$ $I_D = 0.592mA$ or $1.367mA$. $I_D = 0.592mA$ $I_S = 0.592mA$ (1M) $I_D = 0.592mA$ (2M) ; $V_D = 7.04V$ (1M) ; $I_1 = I_2 = 0.25\mu A$ (1M) $V_S = 2.96V$ (1M) ;	8M
2b)	At $V_i = 0$; $\lvert V_{GSN} \rvert = \lvert V_{GSP} \rvert$; $I_{DN} = I_{DP}$; $\therefore V_o = 0V$ Both transistors operate in saturation. $I_{DN} = I_{DP} = \frac{1}{2}(0.5)(2.5 - 0.5)^2 = 1mA$. (3M) At $V_i = 2.5V$; $V_{GSP} = 0$; PMOS is in cut off region & $I_{DP} = 0$; V_o is negative ∴ $V_o \leq V_i$ NMOS operates in Linear region	

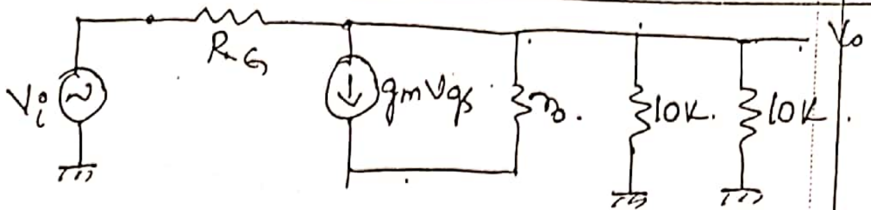
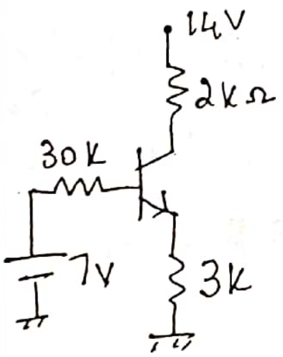
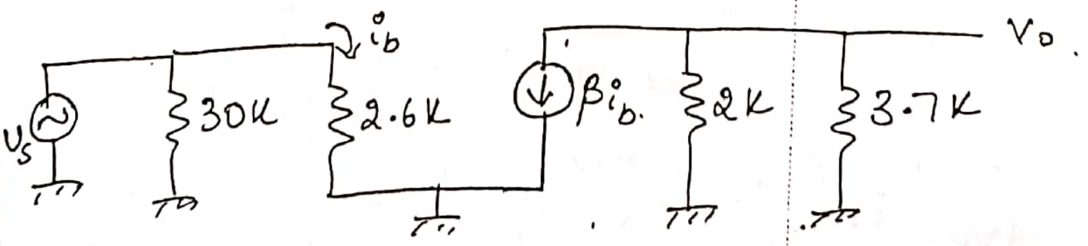
SCHEME AND SOLUTION

SUBJECT CODE: 12EC33 SUBJECT: Analog Microelectronic circuits

Question No		Marks
	$I_D \cong K_n'\left(\frac{W}{L}\right)\left[(V_{GS}-V_t)V_{DS}\right]$; $V_D = V_o = -5I_D$ $I_{D(\text{in mA})} = 0.5\left[(5-0.5)(-5I_D+2.5)\right]$ (5M) $I_D = 0.459\,mA$; $V_o = -2.296\,V$.	8M
3a)	$I_{D(mA)} = \frac{14-V_1}{1.2} \Rightarrow V_1 = 14-1.2I_D$ — (1) (2M) $V_3 = 1.2\,I_{D(mA)}$; $I_{D(mA)} = \frac{0.25}{2}(V_1-V_2-1)^2$ — (2) (2M) $I_{D(mA)} = \frac{0.25}{2}(V_2-V_3-1)^2$ — (3) (2M) Equating (2) & (3) $V_2 = 7V$ From (3) $I_{D(mA)} = \frac{0.25}{2}(14-1.2x-7-1)^2$ $\therefore I_D = 1.82\,mA$; $V_3 = 2.184V$; $V_1 = 11.816V$ (2M)	8M
3b)	$I_{D(mA)} = \frac{K_n'\left(\frac{W}{L}\right)(V_{GS}-V_t)^2}{2}$; $V_G = V_D = 15-10I_D$. $I_D = \frac{0.25}{2}(15-10I_D-1.5)^2$ (2M) $I_D = 1.0589\,mA$; $V_G = V_D = 4.41\,V$. $g_m = K_n'\left(\frac{W}{L}\right)(V_{GS}-V_t) = 0.25\times10^{-3}(4.41-1.5)$ $g_m = 0.73\,mA/V$ (1M) $r_o = \frac{V_A}{I_D} = \frac{50}{1.0589m} = 47.2\,k\Omega$ (1M)	

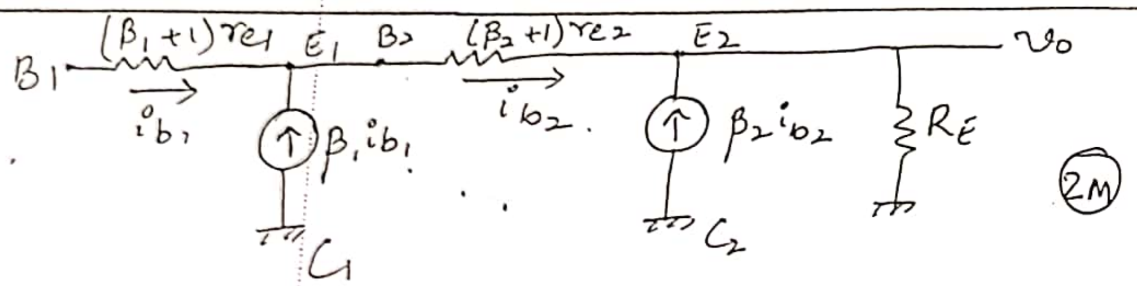
SCHEME AND SOLUTION

SUBJECT CODE: 12EC33 SUBJECT: Analog Micro electronic Circuits

Question No		Marks
		
	$A_V = -g_m(r_o \| 10K \| 10K) = -0.73m(47.2K \| 5K)$ $= -3.3$ (1M)	
	Input Impedance $= \frac{R_G}{1-A_V} = \frac{10M}{1+3.3} = 2.325M\Omega$ (1M)	8M
	Output Impedance $= R_o \| r_o = 10K \| 47.2K = 8.26K\Omega$	
	Largest allowable input Signal $= \frac{V_G}{1-A_V} = 0.348V$ (2M)	
4a)	$R_{th} = 30K\Omega$ $V_{th} = 7V$ (1M)	
		
	$V_{th} = I_B R_B + V_{BE} + I_E R_E$ $7 = 30K I_B + 0.7 + 3K(200) I_B$ $I_B = 10\mu A$; $I_E = 2mA$ (1M) $V_{CE} = 14 - 2K I_C - 3K I_E = 4.02V$ (1M) $r_e = \frac{V_T}{I_E} = \frac{26m}{2m} = 13\Omega$ (1M)	
	$S(I_{co}) = \frac{(\beta+1)(R_{th}+R_E)}{R_B+(\beta+1)R_E} = \frac{200(30K+3K)}{30K+200\times3K} = 10.476$ (1M)	
		
	$\frac{V_o}{V_s} = -\frac{(2K \| 3.7K)}{13} = -99.86$ (1M)	

Question No		Marks
	Input Impedance = $30K \parallel 2.6K = 2.39K\Omega$ (1M) Output Impedance $= 2K\Omega$ (1M)	

4b.



(2M)

$i_{b_2} = (\beta_1+1) i_{b_1}$; $v_o = (\beta_2+1)(\beta_1+1) i_{b_1} R_E$

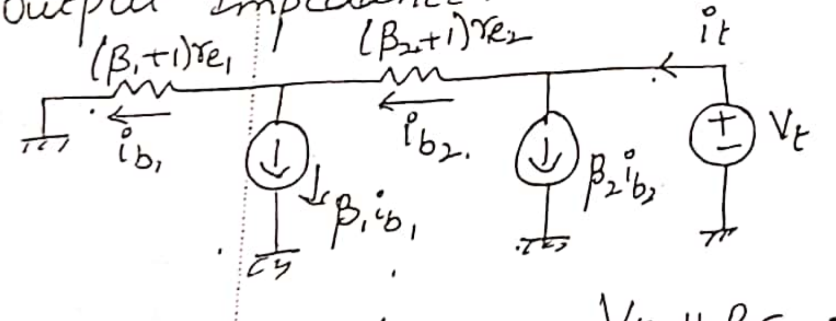
$v_i = (\beta_1+1) r_{e_1} i_{b_1} + (\beta_2+1) r_{e_2} (\beta_1+1) i_{b_1} + R_E(\beta_2+1)(\beta_1+1) i_{b_1}$

$$\frac{v_o}{v_i} = \frac{R_E}{r_{e_2} + R_E + \frac{r_{e1}}{\beta_2+1}}$$ (2M)

Input Impedance $\frac{v_i}{i_{b_1}} = (\beta_1+1) r_{e_1} + (\beta_2+1)(\beta_1+1) r_{e_2} + (\beta_2+1)(\beta_1+1) R_E$

$\cong \beta_2 \beta_1 R_E$ (2M)

Output Impedance.



Output impedance $= \frac{V_t}{i_t} \parallel R_E = R_o$.

$V_t = i_{b_1}[r_{e_1}(\beta_1+1) + (\beta_2+1)(\beta_1+1) r_{e_2}]$

$i_t = i_{b_1} (\beta_2+1)(\beta_1+1) i_{b_1}$

$\frac{V_t}{i_t} = \frac{r_{e_1}}{\beta_2+1} + r_{e_2}$; $\frac{r_{e_2}}{r_{e_1}} = \frac{1}{\beta_2+1} \Rightarrow r_{e_2} = (\beta_2+1) r_{e_1}$ (2M)

$R_o : 2r_{e_2} \parallel R_E \cong 2r_{e_2}$.

8

SCHEME AND SOLUTION

SUBJECT CODE: SUBJECT:

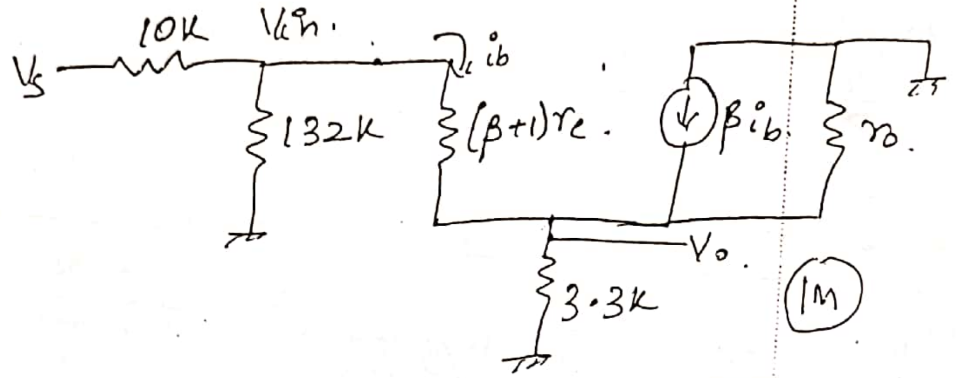
Question No		Marks

5a) $V_{th} = -3V$; $R_{th} = 132K\,\Omega$.

$9 = 3.3K(100)I_B + 0.7 + 132K\,I_B + 3 = 0$

$I_B = 11.47\mu A$; $I_E = 1.147mA$

$r_e = \frac{26m}{1.147m} = 22.67\Omega$ (2M)



(1M)

$\frac{V_o}{V_{in}} = \frac{R_E \| r_o}{r_e + (R_E \| r_o)} = 0.993V$. (1M) 8

$R_i = 132K \| [(\beta+1)(r_e + (R_E \| r_o))] = 91.08K\Omega$ (1M)

$\frac{V_i}{V_s} = \frac{91.08K}{10K + 91.08K} = 0.901$. $A_{VS} = 0.901 \times 0.993 = 0.894$ (1M)

$R_{out} = (R_E \| r_o) \| \left[r_e + \frac{132K \| 10K}{\beta + 1}\right] = 111.2\Omega$ (2M)

5b $R_B = 30K\Omega$; $r_e = 26\Omega$; $r_\pi = 5.2K$; $A_V = -57.28$ (1M)

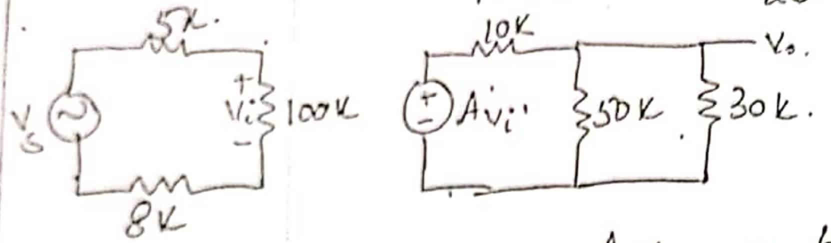
$f_{LC_1} = \frac{1}{2\pi C_1[(R_B \| (r_\pi)) + R_{sig}]} = 29.3\,Hz$ (1M)

$f_{LC_2} = \frac{1}{2\pi C_2(R_C + R_L)} = 26.52\,Hz$ (1M)

$f_{LCE} = \frac{1}{2\pi C_E\left[R_E \| \left(r_e + \frac{R_B \| R_{sig}}{\beta + 1}\right)\right]} = 208.55\,Hz$ (1M)

$C_{in} = C_{be} + C_{bc}(1 - A_V) = 112.58\,pf$ (1M)

SUBJECT CODE: 12EC33 SUBJECT: Analog Microelectronics Circuit

Question No		M
10 b)	Open loop gain $A_{OL} = \frac{100K}{113K} \times 1000 \times \frac{18.75K}{28.75K} = 577$ (1M)  $\beta = 0.2$ (2M) $1 + A_{OL}\beta = 116.4$ (1M) ; $A_{CL} = \frac{A_{OL}}{1 + A_{OL}\beta} = 4.957$ (1M) $R_{i(OL)} = 113 K\Omega$; $R_{if} = 13.15 M\Omega$; $R_{in}(actual) = 13.145 M\Omega$ $R_{o(OL)} = 6.52 K\Omega$; $R_{of} = \frac{R_o}{1 + A\beta} = 56\Omega$; $R_o(actual) = 57.06\Omega$ (1M)	6M
10 c)	Power dissipated in transistors $= P_{idc} - P_{oac} = P_d$. $P_d = \frac{2 V_{cc} V_L(p)}{\pi R_L} - \frac{V_L^2(p)}{2 R_L}$; P_{dmax} is when $V_L(p) = \frac{2V_{cc}}{\pi}$. η at $P_{dmax} = \frac{P_{ac}}{P_{dc}} = 50\%$ (4M)	4M
11 a)	$20 = \frac{A}{1 + A\beta}$ (1M) ; $19.8 = \frac{A/2}{1 + A\beta/2}$ (1M) ; $\frac{20}{19.8} = \frac{2 + A\beta}{1 + A\beta} \Rightarrow A\beta = 98$ $A = 1980$ (1M) ; $\beta = 4.95\%$ (1M)	4M
11 b)	i) $P_{ac} = \frac{V_{cc}^2}{2R} = 6.25W$ (2M) ; ii) $P_{ac} = \frac{V_L(p)^2}{2R} = 4W$ (2M) iii) $P_{dc} = \frac{2 V_{cc} V_L(p)}{\pi R_L} = 6.366W$; $P_d = \frac{P_{dc} - P_{ac}}{2} = 1.18W$ (2M) iv) $P_d = \frac{2}{\pi^2}(P_{o\,max}) = 1.266W$ (2M)	8M
11 c)	T_J — θ_{JC} — T_C — θ_{CS} — T_S — θ_{SA} — T_A i) $\theta_{JC} = (T_J - T_C)/P_d = 2.6°C/W$ (1M) ii) $\frac{T_J - T_S}{P_d} = \theta_{JC} + \theta_{CS}$ (1M) $\Rightarrow T_S = 84°C$ (1M) iii) $\theta_{SA} = \frac{T_S - T_A}{P} = 1.5°C/W$ (1M)	4M

12EC33 / 10EC44

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III / IV Semester B. E. Fast Track Examinations July-16

Electronics and Communication Engineering

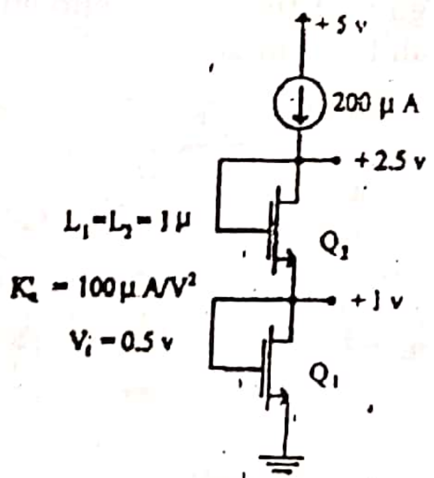
ANALOG MICROELECTRONIC CIRCUITS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	A $PMOS$ device having $\lvert V_t \rvert = 1V$ is biased such that $V_s = 5V$ and $V_G = 2V$. The maximum V_D allowed while the device operates in saturation is = _________.	01
	1.2	An $NMOS$ transistor has $C_{OX} = 10fF/\mu m^2$, $W = 100\mu m$, $L = 0.8\,\mu m$ and $\lvert V_t \rvert = 0.5V$. If its $V_{GS} = 1.5V$, the total charge stored in the channel, for $V_{DS} = 0$, is = _________.	01
	1.3	An $NMOS$ device operating in triode region acts as an $800\,\Omega$ resistor when V_{OV} is $2V$. If the device has to act as a $500\,\Omega$ resistor, then its V_{OV} should be _________.	01
	1.4	The $NMOS$ transistors in the circuit of fig 1.4 are identical in al respects except their gate widths. The required gate widths are $W1 =$ _________ and $W2 =$ _________.  fig.1.4	02
	1.5	A BJT whose VA (Early voltage) is $52V$ at 27°C, has an intrinsic gain of _________.	01

1.6 In the circuit of Fig. 1.6, V_{CE} = ___2___. (assume β to be very large).

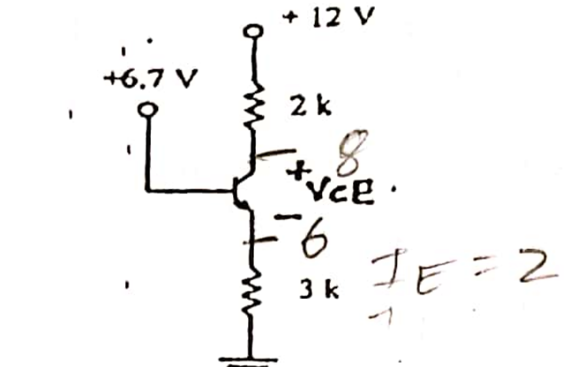


fig.1.6

01

1.7 Assuming the silicon transistor in fig. 1.7, operated in the active region, the input impedance R_{in} = __________.

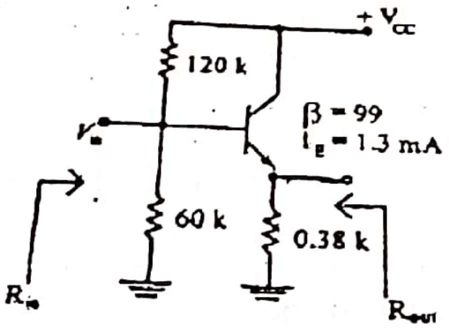


fig.1.7

01

1.8 Assuming the silicon transistor in fig. 1.7, operated in the active region, the output impedance R_{out} = __________. 01

1.9 An $NMOS$ with $V_t = 1, V_S = 1V$ and $V_G = 3V$ and $\lambda = 0.01V^{-1}$ has an intrinsic gain of __________. 01

1.10 The small signal gain of the circuit shown in fig. 1.10 is __________. Assume $\lambda = 0$ for all the transistors.

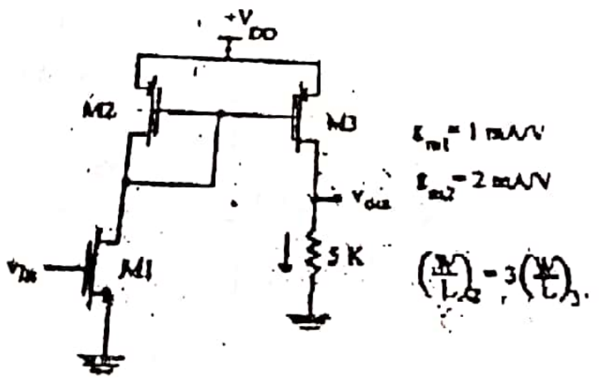


fig.1.10

01

1.11 For the $NMOS$ differential circuit given in fig 1.11, the g_m of $Q1$ is $10mA/V$, the small signal differential gain $\frac{V_o}{V_{id}}$ = ________.

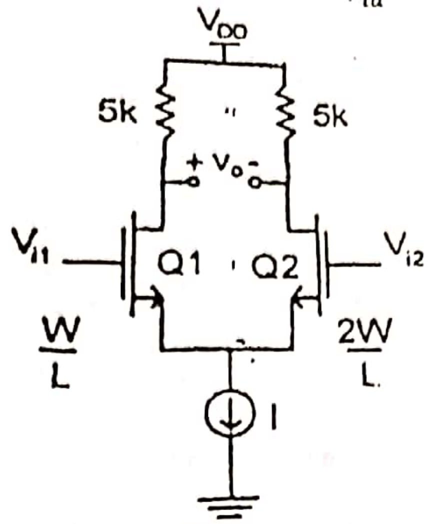


fig.1.11

02

1.12 For the circuit shown in fig. 1.12, the output voltage V_o is = ________.

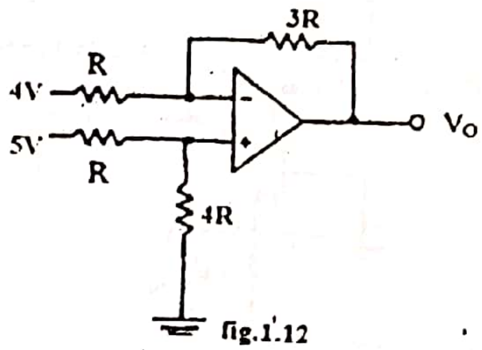


fig.1.12

01

1.13 An opamp has a differential gain of $100dB$ and $CMRR$ of $80dB$. If the differential input is $100\mu V$ and the common input is $100mV$, the output voltage is = ________. 01

1.14 In a voltage follower using an op-amp, whose slew rate is $4V/\mu s$, peak value of the input sinusoidal voltage is $5.3V$. The maximum frequency of the input signal, so that the output is not distorted = ________. 01

1.15 Consider an opamp with finite open loop gain has a unity gain bandwidth of $9MHz$. If it is used as an amplifier with a gain of -5, the $3dB$ frequency of the closed loop amplifier is = ________. 01

1.16 An amplifier has an open loop gain of $40dB$ and an input impedance of $2k\Omega$. If 5% current shunt negative feedback is provided to this amplifier, the input impedance of the closed loop amplifier = ________. 01

1.17 A class B complementary symmetry power amplifier using $BJTs$ is operating from a single supply of $14V$ and driving a loud speaker of 5Ω. The maximum power output possible is = ________. 01

1.18 A power transistor with $T_j(max) = 200^\circ C$ has $\theta_{JA} = 5^\circ C/W$. $P_d(max)$ for the transistor at an ambient temperature of $50^\circ C$ = ________. 01

PART-B

2 a For the circuit shown in Fig.2(a), draw the AC equivalent and determine $\frac{V_o}{V_{in}}$.

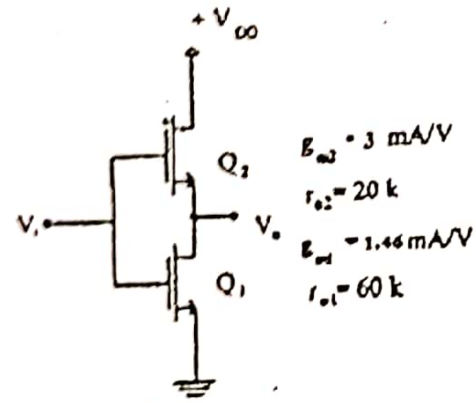


fig.2a

08

b In the circuit of fig 2(b), determine the voltages V_1 and V_2. Assume that the two transistors are identical.

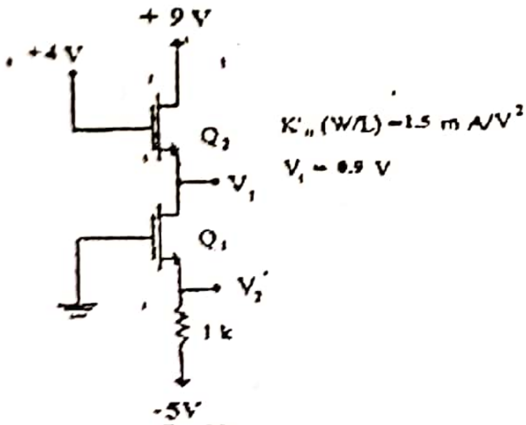


fig.2b

08

OR

3 a Determine the small signal voltage gain, its input impedance and the largest allowable input signal V_i for the amplifier circuit of fig, 3(a). the transistor has $V_t = 1V$, $V_A = 50V$ and $Kn'(W/L) = 0.20mA/V^2$. Assume C_{c1} and C_{c2} are infinite.

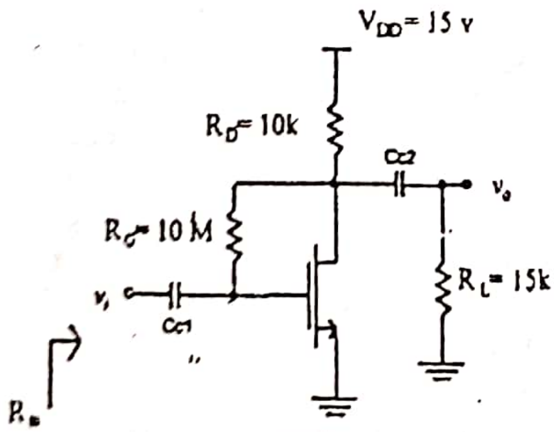


fig.3a

10

b) For the circuit shown in Fig 3(b), determine $V1, V2$ and $V3$. $Kn'(W/L) = 4mA/V^2, V_t = 1V$ and $\lambda = 0$.

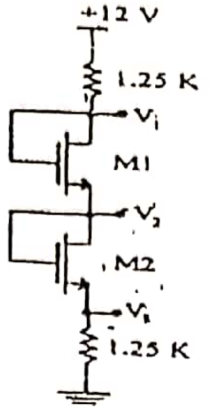


fig.3b

06

4 Design the values of R_c, R_E, R_1 and R_2 in the amplifier circuit of Fig.4 so that the operating point is fixed at $V_{CE} = 5V, I_C = 1.3mA$, with bias stabilization factor $S(I_{CO}) = 16$, and the mid frequency small signal voltage gain $\frac{V_o}{V_{sig}} = -150$. Design the values of C_{c1}, C_{c2} and C_E so that the lower cut off frequency of the amplifier is $100Hz$. Choose C_{c1}, C_{c2} so that the lower cutoff frequency because of C_{c1} and C_{c2} are atleast a decade lower than $100Hz$.

Repeated

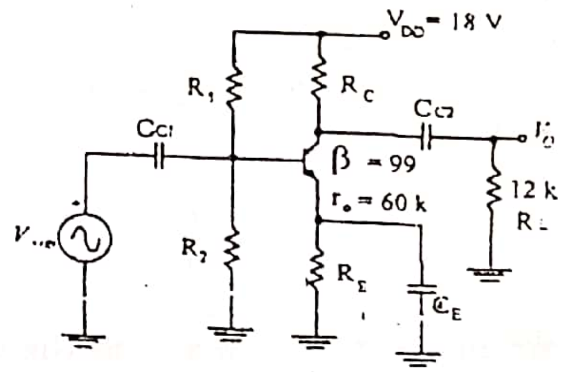


fig.4

16

OR

5 a) In the amplifier circuit of Fig.5(a), determine the lower cutoff frequencies f_{LB}, f_{LE} due to C_B, C_C, C_E respectively and the upper cutoff frequencies f_{hi} and f_{ho} due to the input and output circuits respectively. Also calculate the bandwidth of the amplifier. C_{wi} and C_{wo} are the wiring capacitances at the input and output of the transistor.

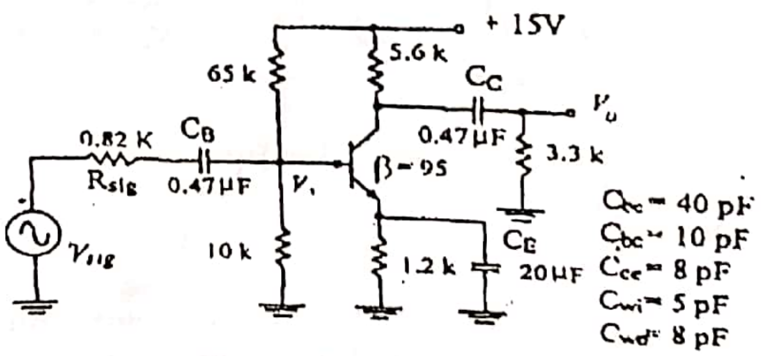
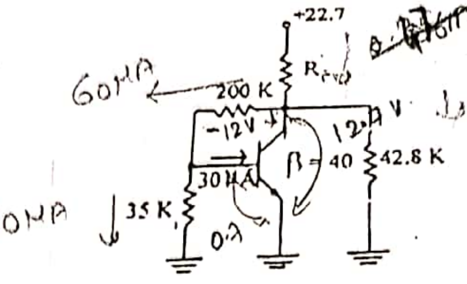


fig.5a

12

b Determine the value of R_C in the circuit of Fig 5(b), if the base current of si transistor is $30\mu A$ and its $\beta = 40$.

1.2m
0.3m
0.06m
1.56m
10
1.56m



60μA
30μA
0.296mA
$i_C = \beta * i_b$
= 1.2mA

Fig.5b

04

6 a A differential amplifier using BJT is shown in Fig. 6(a), determine the differential input resistance R_{id}, differential gain Ad, the common mode input impedance R_{icm}, the worst case common mode gain A_{cm} and $CMRR$, if the drain resistors used are of 4% tolerance. Assume $\beta = 99$, $V_A = 35V$.

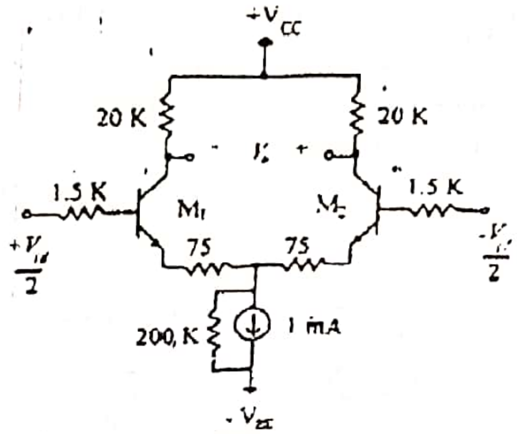


fig.6a

08

b $M1$ and $M2$ are matched transistors in the circuit of Fig 6(b). Find the required value of R and the device dimensions so that the nominal output current $I_o = 100\mu A$, when $V_o = V_{GS}$. It is required that the current source should operate for V_o in the range of 0.2 to V_{DD} and the change in I_o over this range should be limited to 2% of the nominal I_o. Find also the output impedance of the current source.

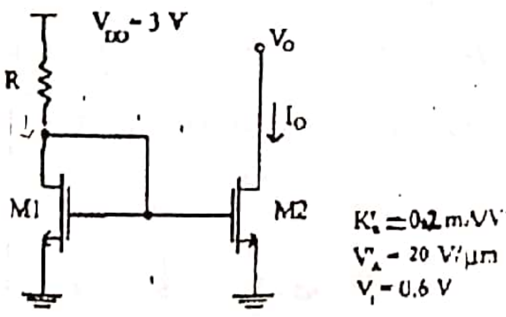


Fig.6(b)

08

OR

7 a Draw the circuit of the Wilson current mirror using bipolar transistors and prove that the output impedance is approximately $\beta r_o/2$. Assume that all transistors are identical.

06

b A current mirror loaded MOS differential amplifier of fig. 7(b) is biased with a current source $I = 1.2mA$. The $NMOS$ device has $V_{ov} = 0.3V$ while that of $PMOS$ device is $0.4V$. $V_{AN} = |V_{AP}| = 9V$. Assume that for all the transistors, $Cgs = 30fF, Cgd = 10fF$ and $Cdb = 0$. In addition to the capacitance introduced by the transistors at the output node, there is capacitance, $C_X = 130fF$. Find the DC value and the frequencies of the poles and zero of the differential voltage gain. **10**

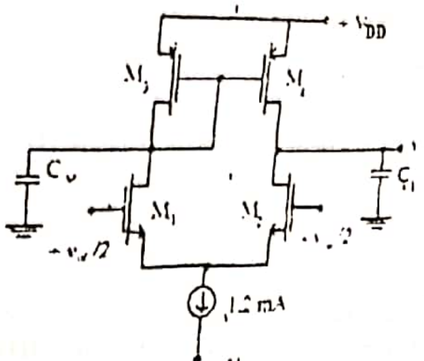


fig.7b

8 a Design an instrumentation amplifier using 3 ideal op-amp, to obtain a differential gain, variable in the range 2 to 100, utilizing a $100k$ pot as a variable resistor. **08**

b Design a difference amplifier circuit using an ideal opamp to realize a differential impedance of $15k\Omega$ and a differential gain of $30dB$. Assuming the resistors used are of 0.5% tolerance, calculate the worst case $CMRR$ of the amplifier. **08**

OR

9 a Design a Schmitt trigger using an ideal opamp to have the upper and lower threshold points as $+4V$ and $+2V$. Assume the opamp is operating with $\pm 12V$. **08**

b Design a second order active low pass filter with upper cut off frequency of $4kHz$ and maximal flat response using Sallen-Key structure. **08**

10 a The circuit of fig 10(a) consists of a differential stage followed by an emitter follower, with negative voltage series feedback, provided by $R1$ and $R2$. Find the values of, A, feedback factor β, A_f, R_{in} and R_{out}. Assume β of the transistors to be 99. **12**

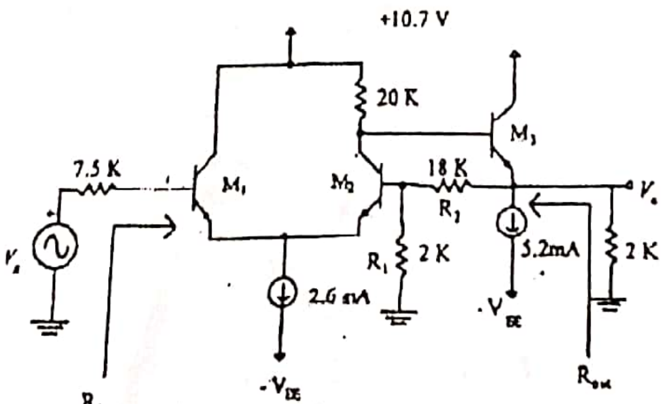


fig.10a

	b	An amplifier has a gain A, input impedance R_i and output impedance R_o. With current shunt negative feedback, obtain the expressions for the input and output impedances of the closed loop amplifier. Assume feedback network is ideal with feedback factor β.	04
		OR	
11	a	A class B output stage is required to deliver a power of $50W$ to a 4Ω load. The power supply should be $5V$ greater than the corresponding peak sine wave output voltage. Determine the supply voltage required, peak current from each supply, total supply power, conversion efficiency and dissipation in each of the transistors.	06
	b	Prove that the conversion efficiency in a class-B power output stage is 50%, when the power dissipation in the transistors is maximum.	04
	c	A silicon transistor operated with a heat sink ($\theta_{CS} = 0.5^\circ C/W$, $\theta_{SA} = 1.5^\circ C/W$) has $\theta_{JC} = 0.5°C/W$. What maximum power can be handled by the transistor at an ambient temperature of 25°C, if $T_j(max) = 175°C$. While dissipating $50W$ of power at $T_A = 30°C$, determine the temperature of the heat sink, case and the junction of the transistor.	06

USN

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Nov/Dec-16

Electronics and Communication Engineering

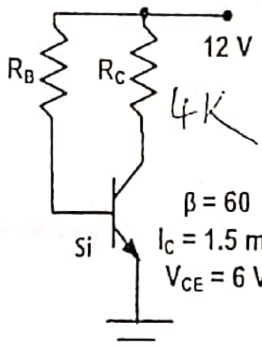
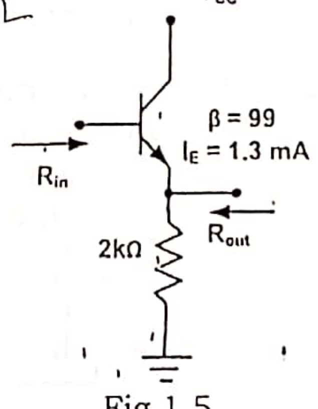
ANALOG MICROELECTRONIC CIRCUITS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	The aspect ratio of a $MOSFET$ is changed by 2 times, while I_D is maintained constant. Determine the change in the transconductance, g_m.	01
	√2		
	1.2	A $MOSFET$ carries a drain current of $1\,mA$ with $V_{DS} = 1V$ in saturation. Determine the change in I_D, if V_{DS}, increases to $1.55\,V$ and $\lambda = 0.1V^{-1}$. Find the output impedance of the device.	02
	55μA 1KΩ		
	1.3	For a $MOSFET$, if $C_{gs} = 80fF$, $C_{gd} = 40fF$ and $g_m = 1mA/V$, the unity gain frequency of the device = ________.	01
	1.32GHz		
	1.4	Determine the values of resistors R_B and R_C in the circuit of Fig 1.4. 452K 4K  Fig 1.4	02
	1.5	In fig.1.5, assuming the transistor to be in active region, the input impedance, R_{in} = ________ and the output impedance, R_{out} = ________. 202KΩ 19.5Ω  Fig 1.5	02

1.6 The value of R_E in Fig.1.6 is equal to ________. Assume β is very large for both the transistors and $V_{BE} = 700\ mV$ @ $I_c = 1mA$.

600Ω

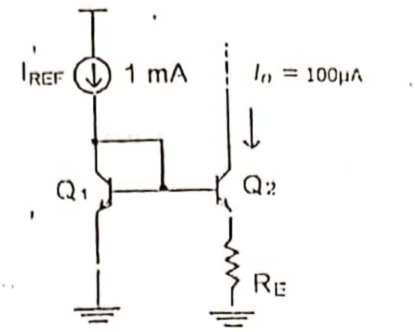


Fig 1.6

1.7 In Fig. 1.7, $I_{EE} = 1.5\ mA$ and the emitter area, A_E of Q_1 is twice that of Q_2. The differential input impedance = ________. Assume $\beta = 99$. 02

7.5KΩ

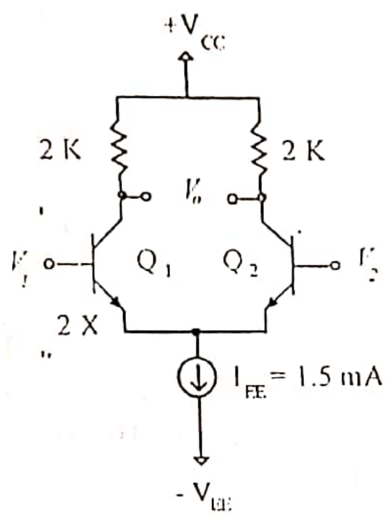


Fig 1.7 02

1.8 Consider the amplifier circuit shown in Fig. 1.8. The opamp is specified to have output saturation voltages of $\pm 13V$, the output current, i_o = ________for $V_p = 1V$.

11mA

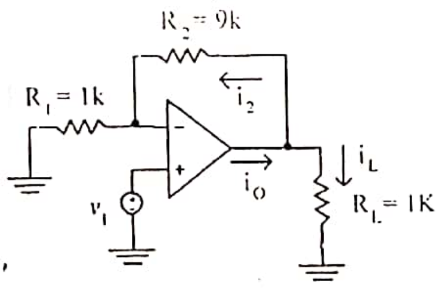


Fig 1.8 02

1.9 An opamp with a unity gain bandwidth of $12MHz$ is used as an amplifier with a closed loop gain of -5. The bandwidth of the closed loop amplifier = 8MHz 01

1.10 Consider an op amp with input resistance $R_i = 50\ K$, output resistance $R_o = 30\ K$ and open-loop gain 299. When such an op-amp is configured as a voltage follower as shown in Fig.1.10, the input impedance, R_{iF} = ________ and the output impedance, R_{oF} = ________.

15MΩ
100Ω

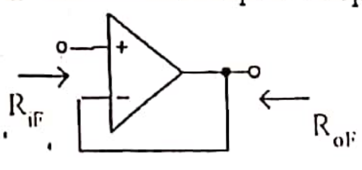


Fig. 1.10

1.11 10KΩ	An amplifier has an open loop gain of $60dB$ and an input impedance of 500 $K\Omega$. If 4.9 % voltage shunt negative is provided to this amplifier, the input impedance of closed loop amplifier = ________.	01
1.12 30W	A power transistor with $T_j(max) = 200^{\circ}C$ has $\theta_{JA} = 5°C/W$, $P_d(max)$ for the transistor at an ambient temperature of $50^{\circ}C$ = ________.	01
1.13 2W	In a class A series-fed power amplifier, suitably biased with a load of 25Ω and $V_{cc} = 20\ V$, the maximum output power = ________.	01

PART-B

2 a In the circuit of Fig.2a, determine V_1 and V_2. The two transistors are identical with $V_t = 1V$, $K'_nW/L = 2mA/V^2$. Assume $\lambda = 0$.

$V_1 = 2.814V$
$V_2 = -2.186$

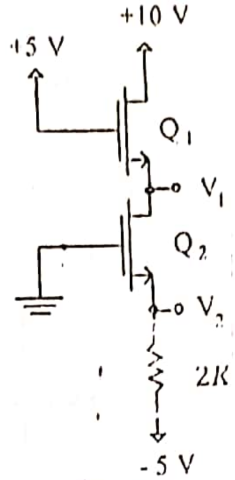


Fig 2a

06

b Design the circuit in Fig 2b so that the transistor operates in saturation with V_D biased 1V from the edge of triode region, with $I_D = 2mA$ and $V_D = 3V$. The device has $|V_t| = 1V$ and $K'_pW/L = 0.5\ mA/V^2$. Assume $10\mu A$ current in the divider.

$V_G = 3V$
$R_1 = 0.7M$
$R_2 = 0.3M$
$R_S = 1.6K\Omega$
$R_D = 1.5K\Omega$

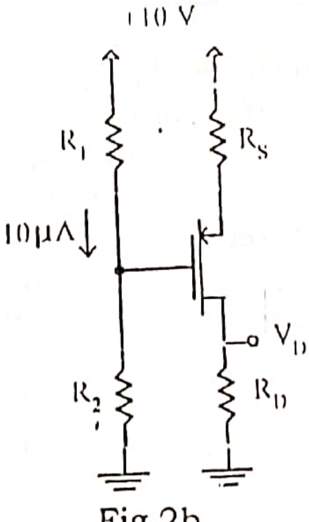


Fig 2b

10

OR

3 a For the circuit shown in Fig 3a, determine V_1, V_2 and V_3. $K'_n W/L = 2.5\ mA/V^2$, $V_t = 1\ V$ and $\lambda = 0$. M_1 and M_2 are identical.

$V_2 = 7V$
$V_1 = 9.69$
$V_3 = 4.31$

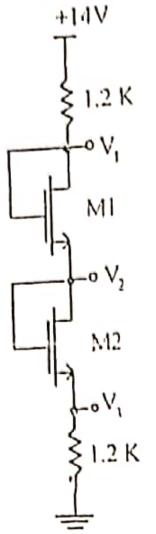


Fig 3a

b In the CS amplifier circuit of fig 3b, determine $I_D, g_m, {}^{v_o}/_{v_i}, R_i, \frac{v_o}{v_s}$ and R_o. Assume $K'_n \frac{W}{L} = 1.2\ {}^{mA}/_{V^2}, V_t = 0.5V$ and $\lambda = 0$. 08

1.867 mA
2.116 mA/V
126k
−12.696
−8.96
6kΩ

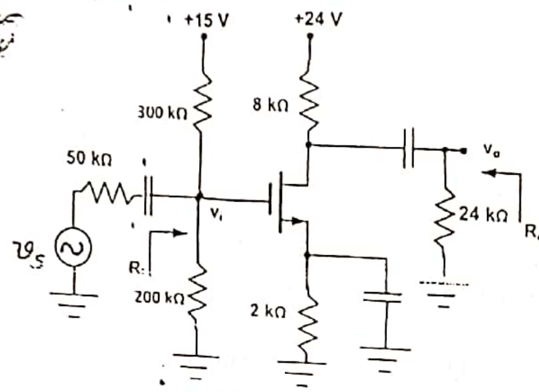


Fig 3b

4 a Design the values of R_C, R_E, R_1 and R_2 in the amplifier circuit of fig. 4a, so that the operating point is fixed at $V_{CE} = 6.3\ V$, $I_C = 2.6\ mA$, with bias stabilization factor, $S(I_{CO}) = 10$, and the mid frequency small signal voltage gain $v_o/v_s = -125$. Design the values of C_{C1}, C_{C2} and C_E so that the lower cut-off frequency of the amplifier is $100Hz$. (Hint: Choose C_{C1} and C_{C2} so that the lower cut-off frequencies because of C_{C1} and C_{C2} are lower than or equal to $10Hz$). 08

$R_C' = 1.25k\Omega$
$R_C = 1.6k\Omega$
$R_E = 3.1k\Omega$
$R_{TH} = 31k\Omega$
$V_{TH} = 9.566V$
$R_1 = 58.33k$
$R_2 = 66.16k$
$C_E = 159.2\mu F$
$C_1 = 16.4\mu F$
$C_2 = 0.97\mu F$

Referred

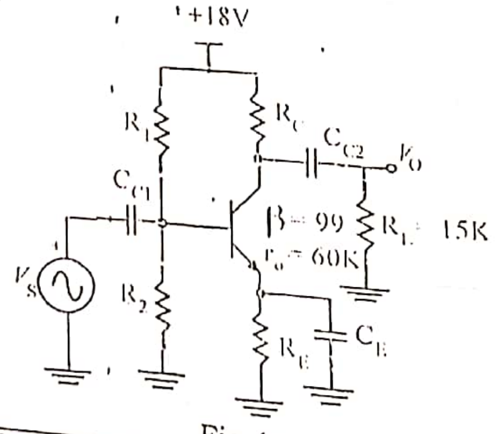
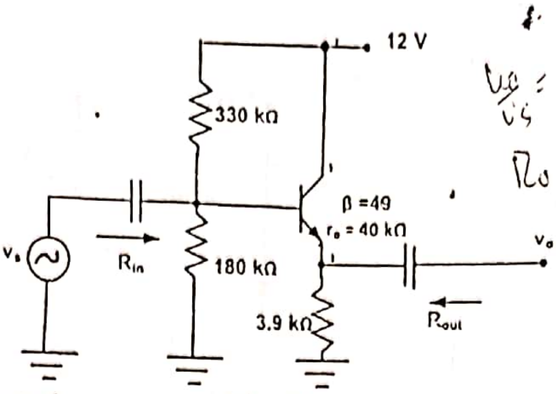


Fig 4a 10

b Determine the operating point of the silicon transistor in the circuit of fig 4b and determine $\frac{v_o}{v_s}$, R_{in} and R_{out}. 06

$11.35\ \mu A$
$0.567\ mA$
$9.8V$



$\frac{v_o}{v_s} = 0.987$
$R_o = 45.85 \,\|\, 3.9k$
$= 45.26\ \Omega$

$r_i = 45.85\ \Omega$
$R_{in} = 116.57k \,\|\, 180k$
$= 70.72\ k\Omega$

Fig 4b

OR

5 a In the amplifier circuit of Fig 5a, determine the lower cut-off frequencies f_{LB}, f_{LC}, f_{LE} due to C_B, C_C and C_E respectively and the upper cut-off frequencies f_{Hi} and f_{Ho} due to the input and output circuit internal capacitances. Also calculate the bandwidth of the amplifier. Assume that the transistor is biased such that $r_e = 39\Omega, C_{be} = 40pF$ and $C_{bc} = 12pF$. Assume $r_0 = \infty$. 10

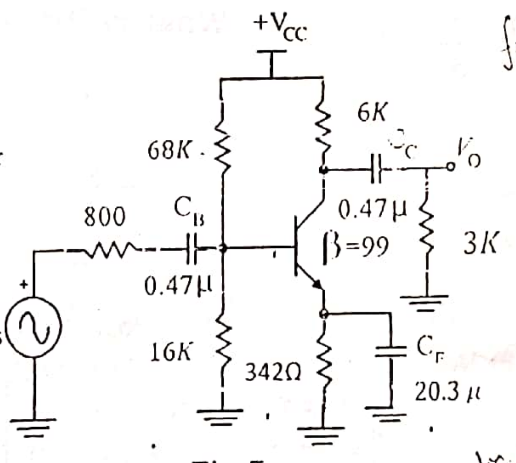
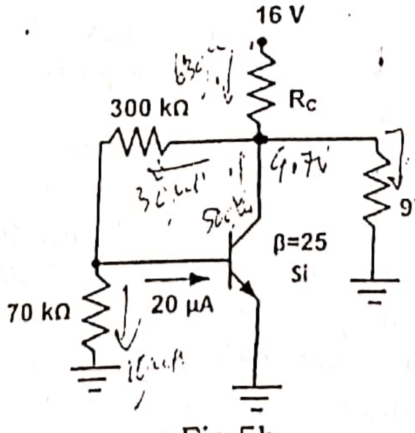


Fig 5a

b In the circuit of fig 5b, determine R_C. 06



1K

Fig 5b

6 a In the Wilson mirror circuit of Fig 6a, determine I_o, and the lowest possible V_o. Also find the output impedance R_{out}, given $V_t = 0.6\,V$, $K'_n = 250\,\frac{\mu A}{V^2}$, $3(\frac{W}{L})_1 = 1.5(\frac{W}{L})_2 = (\frac{W}{L})_3 = 60$ and $V_A = 20V$ for all the transistors.

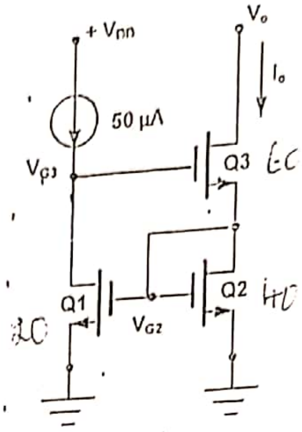


Fig 6a 08

b Design the circuit in Fig 6b to obtain a dc voltage of $+0.2V$ at each of the drains of Q_1 and Q_2 when $V_{G1} = V_{G2} = 0$. Operate all transistors at $V_{OV} = 0.2\,V$ and assume that for the process technology in which the circuit is fabricated, $V_t = 0.5\,V$ and $K'_n = 250\,\frac{\mu A}{V^2}$. Neglect channel-length modulation. Determine the values of R, R_d and the $\frac{W}{L}$ ratios of Q_1, Q_2, Q_3 and Q_4. What is the input common-mode voltage range for the design?

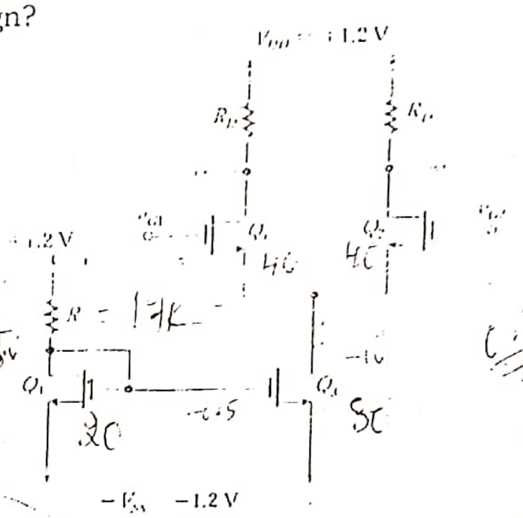


Fig 6b 08

OR

7 a An active loaded MOS differential amplifier of Fig 7a is biased with a current source $I = 0.4\,mA$. The $NMOS$ devices have $V_{OV} = 0.4V$, $V_A = 15\,V$. Assume that for all the transistors, $C_{gd} = 2fF$ and $C_{gs} = 10fF$. The load capacitance at the output node, $C_x = 16fF$. Calculate the values of A_d, A_{cm} and the $CMRR$ at low frequencies. Also determine the pole and zero frequencies of A_d and the dominant pole of $CMRR$, if the capacitance and resistance at node SS is $C_{ss} = 50fF$ and $R_{ss} = 50k\Omega$.

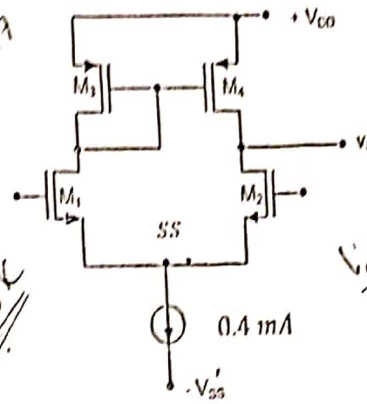


Fig 7a

			08
	b	For the *MOS* differential amplifier circuit shown in Fig. 7b, find V_{OV} and g_m of the devices and A_d, A_{cm} and *CMRR*. Assume the drain resistances are 0.5% tolerance and $\lambda = 0$.	

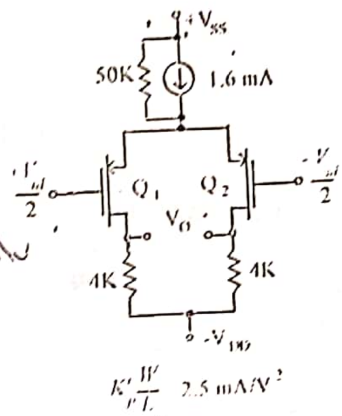


Fig 7b

			08
8	a	Design an instrumentation amplifier, after obtaining the expression for its gain, to realize a differential gain, variable in the range 0.5 to 50, utilizing a 50k pot as a variable resistor.	08
	b	Design a difference amplifier using an ideal op-amp to have a differential input impedance of 10$K\Omega$, differential gain of 28dB and common mode gain of zero. In the designed circuit, if the resistors used are of 1% tolerance, calculate the worst case common mode gain and hence the *CMRR*.	08
		OR	
9	a	Design a Schmitt trigger, using an ideal op amp, to obtain a $UTP = 4V$ and hysteresis $= 6V$. Assume op-amp $V_{SAT} = \pm 12V$.	08
	b	i) Design an inverting amplifier with a closed loop gain of 90 and an input impedance of 1.5 $K\Omega$, using an ideal opamp. ii) If the op amp has an open loop gain of 1547, find the closed loop gain of the amplifier. iii) Design the value of the resistor one can place in parallel with R_1 to restore the closed loop gain to its nominal value.	08

$R_2' = 123.75$ $R_1' = 5.05$ $\frac{R_4'}{R_3'} = \frac{126.25}{4.95}$

$A_{cm} = \frac{123.75}{5.05} - \frac{126.25}{4.95}\left(1+\frac{123.75}{5.05}\right)$

$= 24.505 - 24.543$

$A_{cm} = 0.038$

10 a. In the amplifier circuit of Fig 10a, determine v_o/v_s, R_{in} and R_{out} using feedback methodology. The op amp used has an open loop gain of 800, $R_{id} = 75K\Omega$ and an output impedance of $5K\Omega$.

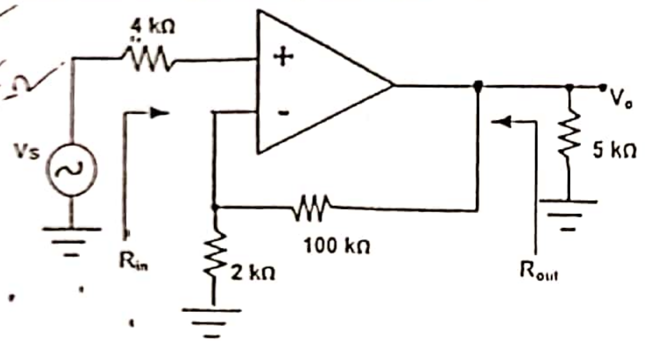


Fig 10 a

08

b Draw the circuit of a complementary symmetry class B power output stage and show that the maximum conversion efficiency is 78.5 %. 04

c Prove that the stability of the gain, A_f, of an amplifier with the negative feedback improves by a factor $(1 + A\beta)$, compared to that of the amplifier without feedback, where A is the open gain and β is the feedback factor. 04

OR

11 a In a complementary symmetry class B power amplifier output stage using a two supply voltages of +16V and $-16V$,

i) What is the maximum power output possible into an 8Ω loud speaker?

ii) If the input voltage is $7\sin\omega t$, what is the power output delivered to the loud speaker?

iii) What is the power dissipation in each of the two transistors, for the input as in (ii) above.

iv) What should be the minimum power dissipation capability of each transistor used in such an output stage. 08

b A transconductance amplifier has an input resistance R_i and output resistance R_o and gain A. If negative current series feedback is provided to such an amplifier with a feedback factor β, derive the expressions for the input and output impedances of the amplifier with feedback. Assume that there are no loading effects of the feedback network. 04

c A silicon power transistor operated with a heat sink ($\theta_{CS} = 1°C/W$, $\theta_{SA} = 2°C/W$) has $\theta_{JC} = 0.5°C/W$. What maximum power can be handled by the transistor at an ambient temperature of 25°C, if $T_j(max) = 175$ °C. 04

R.V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

MODEL QUESTION PAPER

Electronics and Communication Engineering

16EC33: ANALOG MICROELECTRONIC CIRCUITS

Time : 03 Hours — *Maximum Marks : 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Questions 2,7 and 8 of PART B are compulsory
3. Answer questions 3 OR 4 and 5 OR 6

Part- A

Each question carries 1 mark

1

a. For an NMOS transistor, if k'_n W/L = $2mA/V^2$ and I_D= 1mA, its g_m =________.

b. If V_A= 30V and V_{OV}= 0.5V for a MOS transistor, its intrinsic gain =________.

c. A MOS transistor has C_{gs}= 0.1pF and C_{gd}= 25fF. If g_m = 1mA/V, its unity gain frequency =________.

d. A MOS transistor having V_A= 10V is biased such that I_D=1mA and V_{OV}= 2V. If gate and drain are connected together, the incremental resistance between the gate and the source =________.

e. What is 'Early effect' in a Bipolar Junction Transistor?

f. A BJT biased in the active region is used as an amplifier with a collector resistance R_C and an emitter resistance R_E. The small signal voltage gain between base and collector is equal to ________. Assume large β for the transistor and neglect Early effect.

g. A BJT operated in ________ configuration behaves closer to a voltage controlled current source.

h. The output impedance of an emitter follower driven by a voltage source with a source resistance, 10kΩ= ________. Assume β=99 and I_E= 1.3mA for the transistor.

i. The output impedance of a Wilson mirror using identical BJTs, with β=200 and r_o= 50kΩ, =________.

j. For a Wilson mirror, using identical MOSFETs with V_{OV} = 0.5V and V_t = 0.4V, the minimum V_o allowed =________.

k. A MOS differential amplifier with identical transistors has resistive loads of 20kΩ with 2%tolerance. If the current source has a resistance of 100kΩ and the output is taken differentially, the worst case common mode gain = ____.

l. A differential amplifier with identical BJTs with β=199 has a tail current 1mA. The differential input impedance = ________.

m. An op amp has a unity gain bandwidth of 30MHz. If it is used as an amplifier with a closed gain = −5, the bandwidth of the closed loop amplifier = ______.

n. *An op amp has a differential gain of 94dB and CMRR of 100dB. If the two input voltages are 1.00005V and 0.99995V, the output = ________.

o. An op amp has an open loop gain of 999 and an output impedance of 1kΩ. If the op amp is used as a voltage follower, the output impedance of the voltage follower = ________.

p. Current shunt negative feedback is used to ________ the input impedance and ________ the output impedance.

q. The distortions of an amplifier in the open loop and closed loop are respectively 22% and 2%. If the open loop gain is 10^3, the feedback factor = ______.

r. In a series fed class A power amplifier, for obtaining a power output of 4W into a load of 8Ω, the minimum supply voltage needed =_______.

s. A single supply of 12V is used for a complementary symmetry class B power amplifier. If the load is 8Ω, the maximum power output possible =_______.

t. What is Channel Length Modulation in a MOS transistor

Part- B

2. a) In the circuit of fig2a, determine V_1, V_2 and V_3. Q_1 and Q_2 are identical with $k'_n W/L = 2mA/V^2$ and $V_t = 1V$. 08M

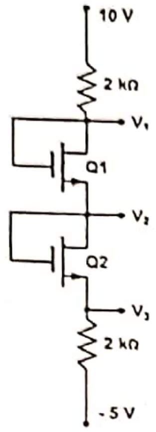


fig 2a

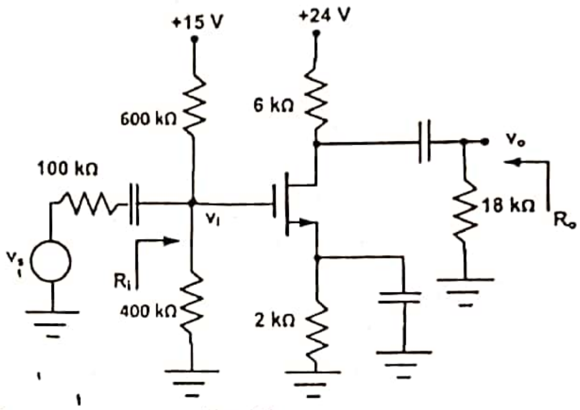


fig 2b

b) The CS amplifier circuit shown in fig2b uses a MOS transistor with $V_t = 0.5V$, $\lambda = 0$ and $k'_n W/L = 1.2mA/V^2$. Determine I_D, g_m, v_o/v_i, R_i, v_o/v_s and R_o. 08M

3. a) Determine V_{CC} in the circuit of fig3a. Assume Silicon transistor. 06M

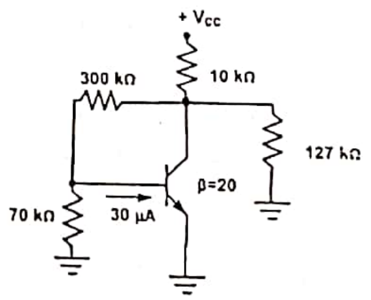


fig 3a

b) Design a CE amplifier circuit using a Si NPN transistor with voltage divider biasing to have the operating point at V_{CE}=5.8V, I_E=1.5mA with $S(I_{CO})$= 20 and the small signal voltage gain = -125. Assume V_{CC}= 16V and transistor β=79 and r_o= 20kΩ. For the designed circuit, determine the input and output impedances. 10M

OR

4. a) Draw the small signal equivalent of a Darlington pair & obtain the expressions for the voltage gain and input and output impedances. 08M

b) In the amplifier circuit of fig4b, determine the operating point of the Si transistor, r_e, v_o/v_i, R_{in}, v_o/v_s and R_{out}. Assume β=49 and r_o= 50kΩ.

08M

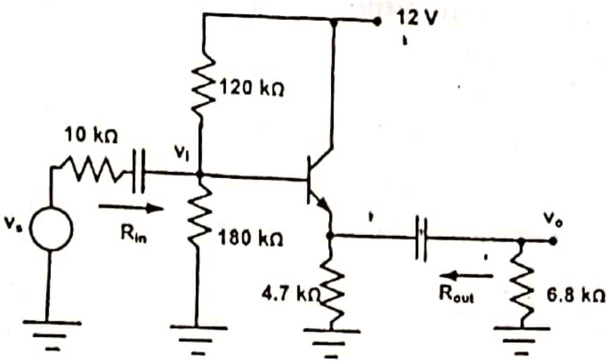


fig 4b

5.a) In the Wilson mirror circuit of fig5a, determine I_o, V_{G1}, V_{G3} and the lowest possible V_o. Find the values of g_m and r_o for all transistors and also the output impedance, R_{out}. 08M

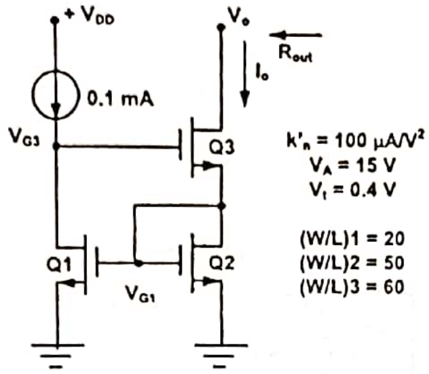


fig 5a

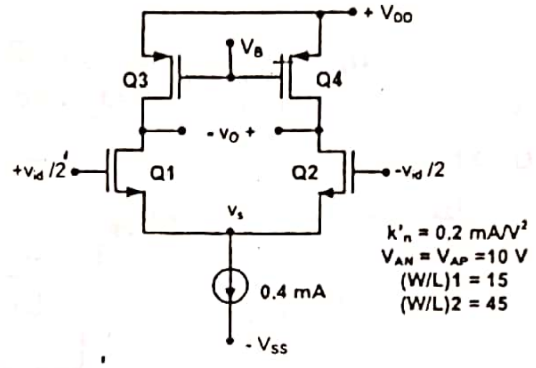
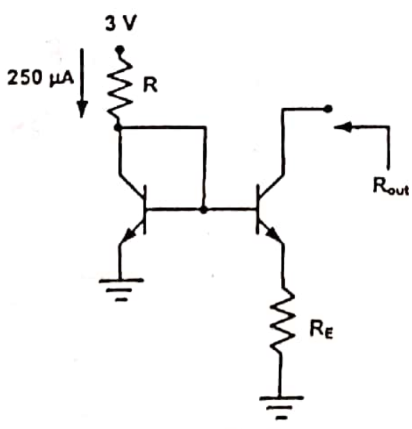


fig 5b

b) For the differential amplifier circuit shown in fig5b, determine I_{D1}, I_{D2}, g_{m1}, g_{m2}, V_s, r_{o1}, r_{o2}, r_{o3}, r_{o4} and the differential gain v_o/v_{id}. 08M

OR

6. a) Calculate the values of R and R_E in the Widlar mirror circuit of fig6a for an output current of 25μA. Assume large β for the transistors and $V_{BE} = 700\text{mV}$ @ $I_C = 1\text{mA}$. For the designed circuit, calculate the output impedance, if $V_A = 20\text{V}$ and $\beta = 99$. 08M



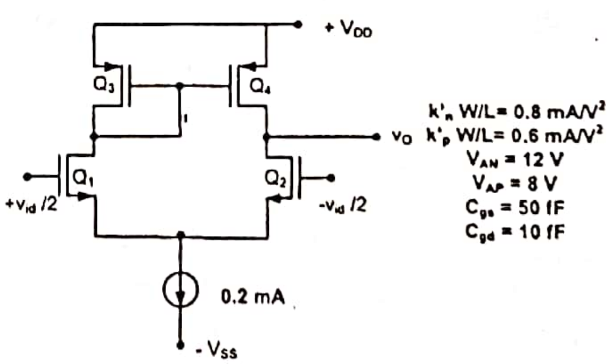


fig 6a fig 6b

b) In the active loaded differential amplifier circuit shown in fig6b, determine the low frequency differential gain, A_d, the effective capacitances C_m and C_o at the mirror and the output nodes respectively. The load capacitance at the output is 70fF. Determine the pole and zero frequencies of

A_d and the zero frequency of the common mode gain. Assume $R_{SS} = 100k\Omega$ and $C_{SS} = 0.1pF$. Also plot the frequency response of CMRR. 08M

7. a) Design an amplifier to have a gain of −40 and an input impedance of 5kΩ, using an op amp which is ideal in all respects except that it has a finite gain of 600. In the designed circuit, what would be the gain if the op amp is replaced by an ideal one? What would be the modification needed to obtain the original gain with the same input impedance with an ideal op amp? 08M

b) Design an instrumentation amplifier using three ideal op amps, after obtaining the expression for the gain, to have a variable gain from 0.5 to 50 using a 50kΩ pot. 08M

8 .a) With the help of a neat schematic, prove that the input and output impedances of an amplifier increase by (1+Aβ), with current-series negative feedback. A is the open loop gain of the amplifier and β is the feedback factor. 08M

b) In the circuit of fig8b, determine v_o/v_s, R_{in} and R_{out} using feedback methodology. The op amp has an open loop gain of 1000, input impedance of 50kΩ and output impedance of 2kΩ. 08M

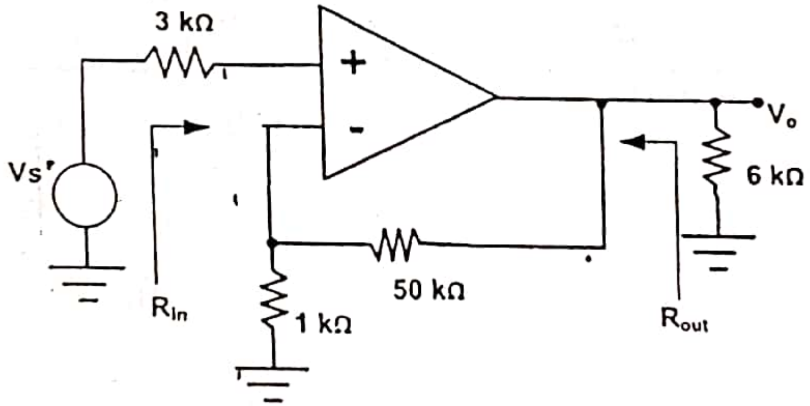


Fig8b

USN | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III Semester B. E. Examinations Nov/Dec-17

Electronics and Communication Engineering

ANALOG MICROELECTRONIC CIRCUITS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	A MOS transistor has its $C_{gs} = 0.4\,pF$, $C_{gd} = 0.1\,pF$. To obtain a unity gain frequency of 200 MHz for the device, its g_m should be = ________.	01
	1.2	A $PMOS$ transistor having $V_t = -0.5\,V$ is biased such that $V_s = 3V$ and $V_G = 2V$. The maximum V_D allowed so that the transistor operates in saturation = ________.	01
	1.3	An $NMOS$ transistor is having $V_t = 0.8\,V$ and $\lambda = 0.02\,V^{-1}$. If its $V_{GS} = 1\,V$, its intrinsic gain = ________.	01
	1.4	For a MOS transistor, if W/L is increased 9 times while maintaining its $I_D$constant, its g_m will increase by a factor of ________.	01
	1.5	In the circuit of fig 1.5, assuming that the transistor is operating in the active region, the incremental resistance between the terminals A and B = ________.	01

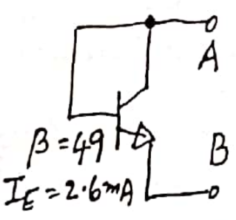


fig 1.5

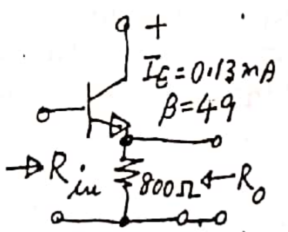


fig 1.6

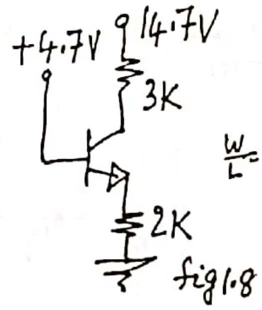


fig 1.8

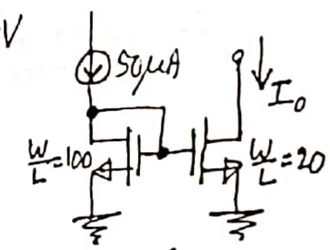


fig 1.9

	1.6	The input impedance, R_{in} in the circuit of fig 1.6 = ________.	01
	1.7	The output impedance, R_o in the circuit of fig 1.6 = ________.	01
	1.8	In the circuit of fig 1.8, assuming the transistor has very large β, V_{CE} = ________.	01
	1.9	In the current mirror circuit of figure 1.9, I_o = ________.	01
	1.10	The output impedance R_o in the circuit of figure 1.10 = ________.	01

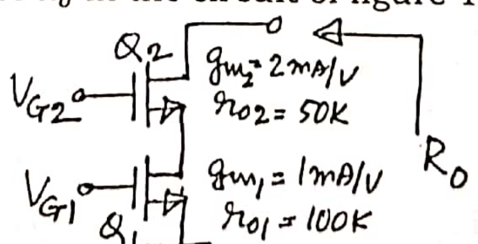
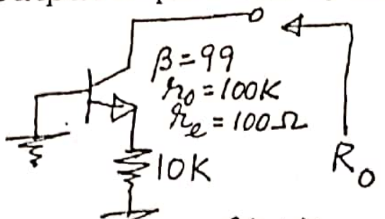
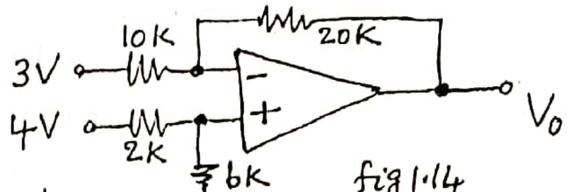
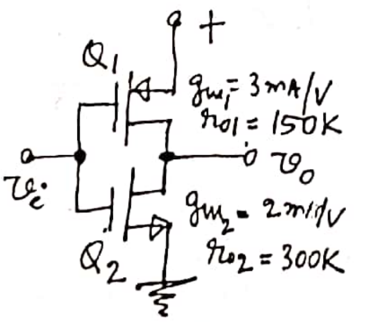
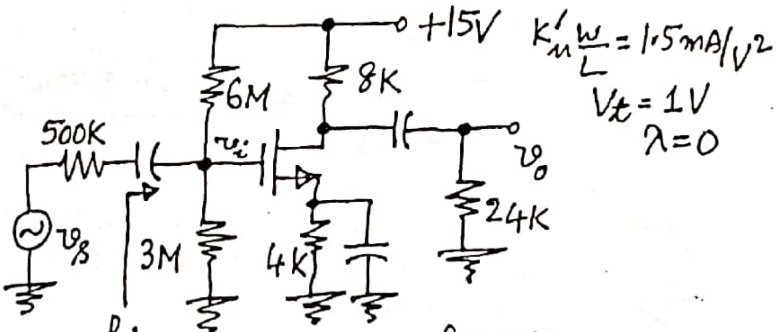


fig 1.10

	1.11	The output impedance R_o in the circuit of fig 1.11 = ________.  fig1.11  fig 1.14	01
	1.12	A MOS differential pair uses a current source with $R_{SS} = 100\ K\Omega$. Assuming for the transistor, $g_m = 4\ mA/V$, if the output is single ended, $CMRR$ = ________.	01
	1.13	A differential amplifier has a differential gain of $80\ dB$ and $CMRR = 86\ dB$. If the differential input is $10\ \mu V$ and the commom mode input is $25\ mV$, the output voltage = ________.	01
	1.14	The output voltage V_o, in the circuit of fig 1.14 = ________.	01
	1.15	An op amp has its $f_t = 10\ MHz$. The 3-dB frequency of a closed loop amplifier with a gain of −4, using the above op amp = ________.	01
	1.16	An op amp has an open loop gain of 500 and $R_{in} = 30\ K\Omega$. If this op amp is used as a voltage follower, the input impedance = ________.	01
	1.17	A Class-B complementary symmetry power output stage is operated with a single supply of $16\ V$ and delivers power to a $8\ \Omega$ loud speaker. The maximum power output possible = ________.	01
	1.18	A power transistor has its $T_j(Max) = 175\ °C$, $\theta_{JC} = \theta_{CS} = \theta_{SA} = 0.8\ °C/W$. The maximum power that the transistor can dissipate at $T_A = 55\ °C$ is ________.	01
	1.19	Current shunt feedback is provided to an amplifier to ________ its output impedance.	01
	1.20	Voltage shunt feedback is provided to an amplifier to ________ its input impedance.	01

PART-B

2	a	Draw the small signal equivalent of the circuit shown in figure 2a and determine the voltage gain, v_o/v_i, the output impedance, assuming the MOS transistors are operating in saturation.  fig2(a)  fig 2(b)	06
	b	In the CS amplifier circuit of fig 2b, determine: i) I_D ii) voltage gain, v_o/v_i iii) input impedance, R_i iv) v_o/v_s v) The maximum peak value of the signal v_s which can be amplified without any distortion.	10

3 a Design the values R_c, R_E, R_1 and R_2 in the CE amplifier circuit of fig 3a, to obtain $V_{CE} = 7V$, $I_E = 1.5\,mA$, $S(I_{CO}) = 15$ and the small signal voltage gain $\frac{v_o}{v_i} = -100$. The transistor has $\beta = 99$ and $r_o = 30\,K\Omega$. repeat — 08

b If figure 3b, determine the operating point of the Si PNP transistor, the small signal voltage gains v_o/v_i, v_o/v_s, the input impedance, R_i and the output impedance R_o. The transistor has $\beta = 99$ and $r_o = 30\,K\Omega$.

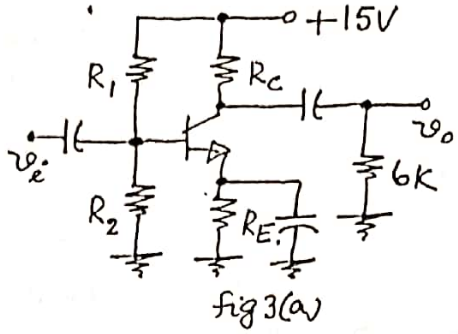


fig 3(a)

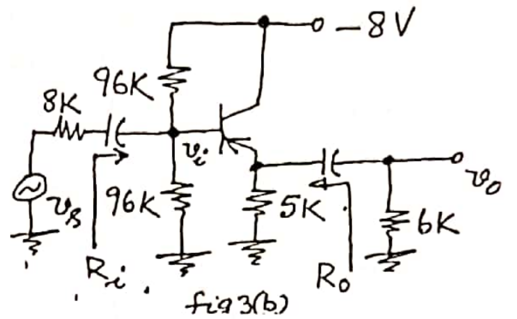


fig 3(b) — 08

OR

4 a Determine the value of R_c in the circuit of fig 4a, if the base current of the Si transistor is $40\,\mu A$ and its $\beta = 20$.

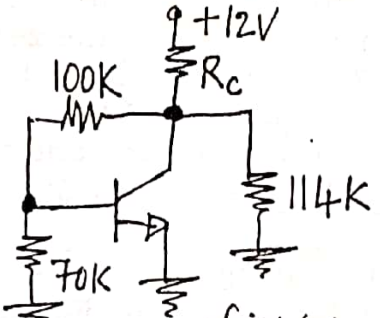


fig 4(a)

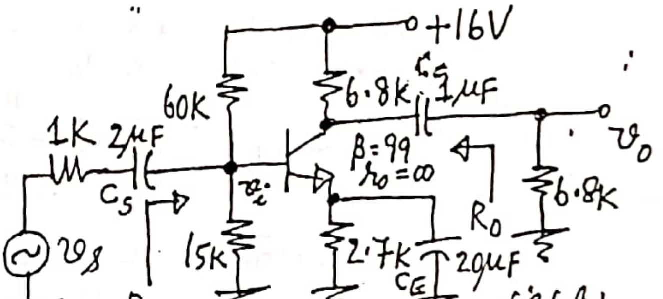


fig 4(b) — 04

b In the amplifier circuit of fig 4b, determine mid frequency gains v_o/v_i and v_o/v_s, the input impedance, R_i and the output impedance R_o, the lower cutoff frequencies f_{LS}, f_{LC} and f_{LE} due to C_S, C_C and C_E respectively and the upper cut off frequencies, f_{hi} and f_{ho} due to the input and output capacitances respectively. Assume $\beta = 99$, $C_{be} = 50\,pF$ and $C_{bc} = 5\,pF$ for the transistor. — 12

5 a In the Wilson mirror circuit of figure 5a, determine I_o, V_{G2}, V_{G3} and the lowest possible V_o. Find the values of g_m and r_o of Q_1, Q_2 and Q_3 and also the output impedance, R_o. Assume $V_t = 0.6\,V$, $k_n' = 200\,\mu A/V^2$, $\left(\frac{W}{L}\right)_1 = 40$, $\left(\frac{W}{L}\right)_2 = 20$, $\left(\frac{W}{L}\right)_3 = 30$ and $V_A = 12\,V$ for all the transistors.

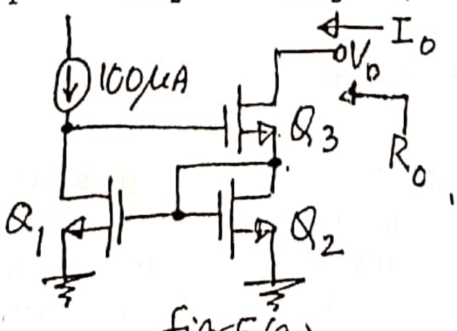


fig 5(a)

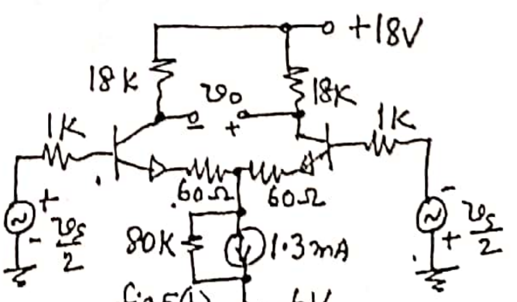


fig 5(b) — 08

	b	For the differential amplifier circuit shown in fig 5b, find the common mode input voltage range, 'the input impedance, R_{id}, the differential gain, v_o/v_s, the common mode input impedance R_{icm}, the common mode gain, A_{cm} and the $CMRR$, if the drain resistors are of 0.5 % tolerance. . Assume $\beta = 99$ and $V_A = 100\ V$ for the transistors and $V_{CS}(Min) = 1V$.	08
		OR	
6	a	In the Widlar's mirror circuit of figure 6a, determine R and R_E, assuming β is very high and $V_{BE} = 700\ mV\ @\ I_C = 1\ mA$. Find the output impedance of the current source, R_o, if $\beta = 120$ and $V_A = 10V$.	06

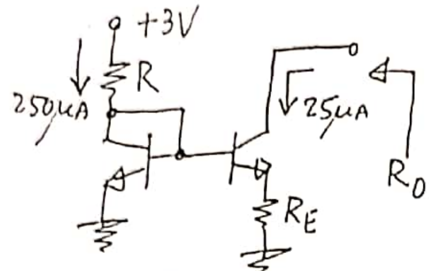


fig 6a

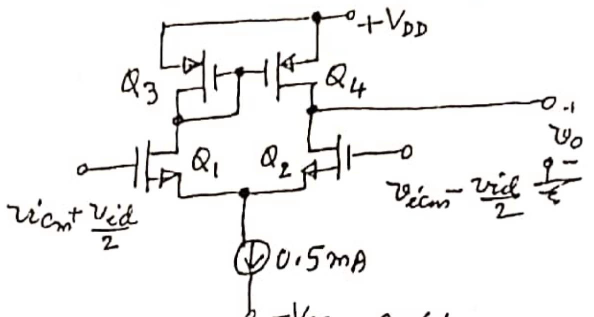


fig6b

	b	In the MOS differential amplifier circuit of fig 6b, find the mid frequency values of the differential and common mode gains. Calculate the pole and zero frequencies of A_d and the zero frequency of A_{cm}. Assume$(V_{OV})_N = 0.25\ V$, . $(V_{OV})_P = 0.5\ V$, $V_{AN} = 10\ V$ and $\lvert V_{AP}\rvert = 8V$. Assume for all the transistors, $C_{gs} = 20\ fF, C_{gd} = 5\ fF$ and $C_{db} = 3\ fF$. There is in addition a load capacitance of 22 fF at the output and $R_{SS} = 60\ K$ and $C_{SS} = 50\ fF$.	10
7	a	Design an instrumentation amplifier, using three ideal op amps, to realize a differential gain, variable in the range 1 to 80, utilizing a 50 K pot as a variable resistor.	08
	b	Design a difference amplifier using an ideal op amp to have a differential input impedance of 12 $k\Omega$, a differential gain of 32.04 dB and zero common mode gain. In the designed circuit, if the resistors used are of 2 % tolerance, determine. the worst case common mode gain and $CMRR$.	08
8	a	The op amp in the circuit of fig 8a has an input impedance of 150 $k\Omega$ and output impedance of 4 $k\Omega$ and an open loop gain of 1200. Determine v_o/v_s, R_{in} and R_{out} using feedback theory.	10

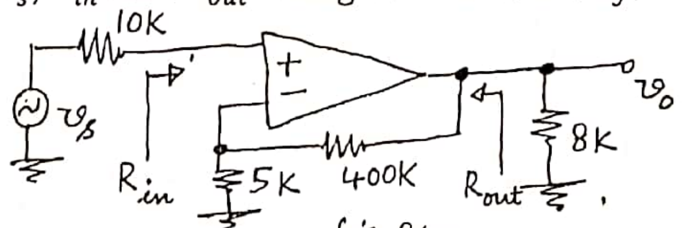


fig 8(a)

	b	A class B complementary symmetry power output stage operates from +16V and−16 V supplies. If the load is 16 Ω and the input is 12 sin 6280 t, determine the output power, input power, efficiency, dissipation in each transistor. Also find the minimum power dissipation capability required for the transistors, so that they operate safely in this circuit.	06

3	a	Design the values R_C, R_E, R_1 and R_2 in the CE amplifier circuit of fig 3a, to obtain $V_{CE} = 7V$, $I_E = 1.5\ mA$, $S(I_{CO}) = 15$ and the small signal voltage gain $\frac{v_o}{v_i} = -100$. The transistor has $\beta = 99$ and $r_o = 30\ K\Omega$. repeat	08
	b	If figure 3b, determine the operating point of the Si PNP transistor, the small signal voltage gains v_o/v_i, v_o/v_s, the input impedance, R_i and the output impedance R_o. The transistor has $\beta = 99$ and $r_o = 30\ K\Omega$.	08

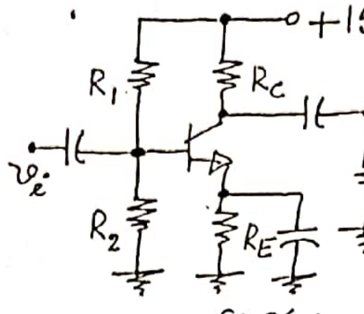


fig 3(a)

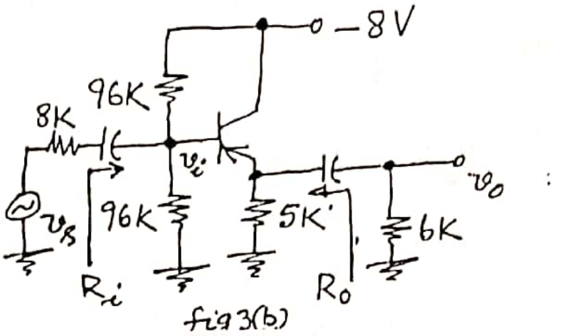


fig 3(b)

OR

4	a	Determine the value of R_c in the circuit of fig 4a, if the base current of the Si transistor is 40 μA and its $\beta = 20$.	04

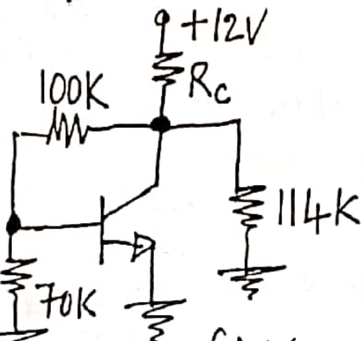


fig4(a)

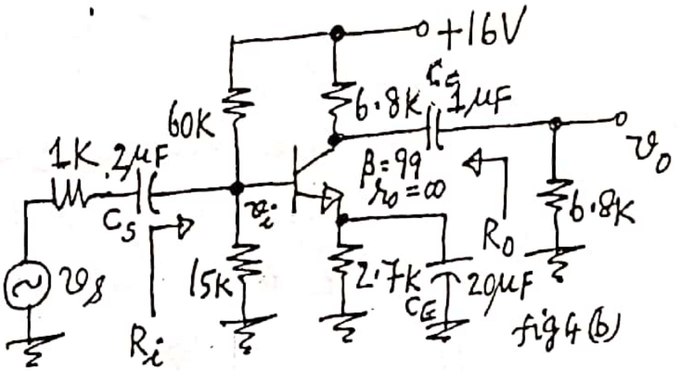


fig4(b)

	b	In the amplifier circuit of fig 4b, determine mid frequency gains v_o/v_i and v_o/v_s, the input impedance, R_i and the output impedance R_o, the lower cutoff frequencies f_{LS}, f_{LC} and f_{LE} due to C_S, C_C and C_E respectively and the upper cut off frequencies, f_{hi} and f_{ho} due to the input and output capacitances respectively. Assume $\beta = 99$, $C_{be} = 50\ pF$ and $C_{bc} = 5\ pF$ for the transistor.	12
5	a	In the Wilson mirror circuit of figure 5a, determine I_o, V_{G2}, V_{G3} and the lowest possible V_o. Find the values of g_m and r_o of Q_1, Q_2 and Q_3 and also the output impedance, R_o. Assume $V_t = 0.6\ V$, $k_n' = 200\ \mu A/V^2$, $\left(\frac{W}{L}\right)_1 = 40$, $\left(\frac{W}{L}\right)_2 = 20$, $\left(\frac{W}{L}\right)_3 = 30$ and $V_A = 12\ V$ for all the transistors.	08

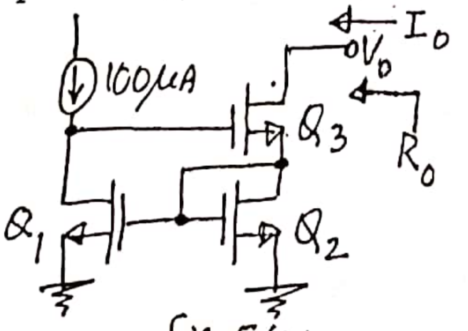


fig 5(a)

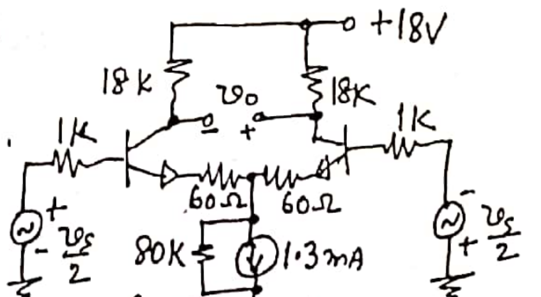
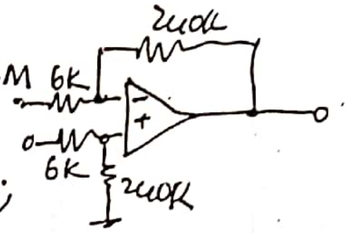
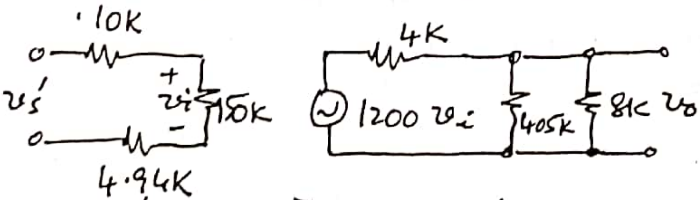
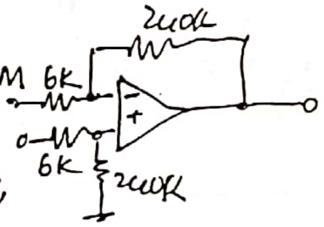
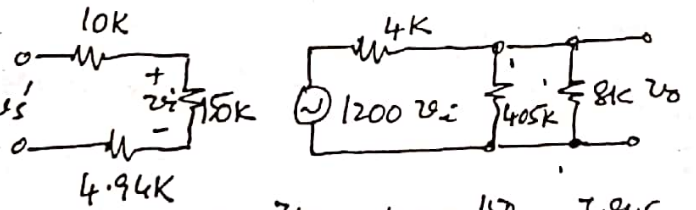


fig 5(b)

Question No		Marks
6 (a)	$R_E = 2.4K$; $R = 9.344K$; $R_E' = 2.355K\Omega$; $\Downarrow 2M$ $\Downarrow 1M$ $\Downarrow 1M$ $R_0 = 2.355K + 400K + \frac{400K}{1.04K} \times 2.355K = 1.3M\Omega$ $\Downarrow 2M$	06M
(b)	$g_{m1} = g_{m2} = 2mA/V$; $r_{o1} = r_{o2} = 40K$; $R_0 = 17.77K$; } $\Rightarrow 2M$ $g_{m3} = g_{m4} = 1mA/V$; $r_{o3} = r_{o4} = 32K$; $A_d = g_{m1}R_0 = 35.55$; $\Downarrow 1M$ $A_{cm} = \frac{32K}{33 \times 120K} = \frac{1}{123.75}$ $\Downarrow 1M$; $CMRR = 4.4 \times 10^3 \simeq 73dB$ $\Downarrow 1M$ $C_m = 51fF$ $\Downarrow 1M$; $C_L = 38fF$ $\Downarrow 1M$; $f_{p1} = \frac{1}{2\pi C_L R_0} = 235.8MHz$ $\Downarrow 1M$; $f_{p2} = \frac{g_{m3}}{2\pi C_m} = 3.122GHz$ $\Downarrow 1M$ $f_z(A_d) = \frac{2g_{m3}}{2\pi C_m} = 6.244GHz$; $f_z(A_{cm}) = \frac{1}{2\pi R_{SS}C_{SS}} = 53MHz$ $\Downarrow 1M$	10M
7(a)	Instrumentation amplifier circuit $\Rightarrow 2M$ Expression for gain $= \left(1 + \frac{2R_2}{R_1}\right)\frac{R_4}{R_3} \Rightarrow 1M$ Let $\frac{R_4}{R_3} = \frac{1}{4}$; then $\left(1 + \frac{2R_2}{R_1}\right) \Rightarrow 4$ to 320 ; $R_1 = R_{1f} + 50K\,pot \Rightarrow 1M$ $\left(1 + \frac{2R_2}{R_{1f}}\right) = 320$ $\Downarrow 1M$ and $\left(1 + \frac{2R_2}{R_{1f} + 50K}\right) = 4$ $\Downarrow 1M$; $R_{1f} = 474.7\Omega \Rightarrow 1M$ $R_2 = 75.71K\Omega \Rightarrow 1M$	08M
(b)	Difference amplifier circuit $\Rightarrow 2M$  Design of 4 resistors $\Rightarrow 1M$ $\frac{R_2'}{R_1'}$ and $\frac{R_4'}{R_3'}$ worst case values $\Rightarrow 1M$; $A_d \Rightarrow 1M$; $A_{cm} \Rightarrow 2M$ $CMRR \Rightarrow 1M$; $\frac{R_2'}{R_1'} : 41.6326$ $A_d = 41.592$; $A_{cm} = 0.0812$; $\frac{R_4'}{R_3'} = 38.4313$ $CMRR = 512.2 \simeq 54dB$	08M
8(a)	 Loading effects $\Rightarrow 1M$ @ output 405K in ll @ input 4.9K in series $\beta = \frac{5}{405} \Rightarrow 1M$ $A = \frac{v_o}{v_s} = 1200 \times \frac{150}{164.94} \times \frac{7.845}{11.845} = 722.78 \Rightarrow 2M$ $R_i = 164.94K$ $\Downarrow 1M$; $R_o = 2.649K$ $\Downarrow 1M$; $(1 + A\beta) = 9.923 \Rightarrow 1M$ $A_f = 72.84$; $R_{if} = 1.636M\Omega$; $R_{of} = 267\Omega$ } $R_{in} = 1.626M\Omega$; $R_{out} = 276\Omega$ } $3M$	10M
(b)	$P_o = 4.5W \Rightarrow 1M$; $\eta = 58.9\% \Rightarrow 1M$ $P_{in} = 7.64W \Rightarrow 2M$; $P_d = 1.57W$ in each tr $\Rightarrow 1M$ Minimum Power dissipation capability for each transistor $= \frac{2}{\pi^2} P_{o(max)} = 1.623$ watts $\Rightarrow 1M$	06M

SCHEME AND SOLUTION NOV/DEC-17

SUBJECT CODE: 16EC33 SUBJECT: Analog Microelectronic Circuits UG

Question No		Marks
6 (a)	$R_E = 2.4K$; $R = 9.344K$; $R_E' = 2.355K\Omega$; ⇓ 2M ⇓ 1M ⇓ 1M $R_0 = 2.355K + 400K + \frac{400K}{1.04K} \times 2.355K = 1.3M\Omega$ ⇓ 2M	06M
(b)	$g_{m1} = g_{m2} = 2mA/V$; $r_{o1} = r_{o2} = 40K$; $g_{m3} = g_{m4} = 1mA/V$; $r_{o3} = r_{o4} = 32K$; $R_0 = 17.77K$; } ⇒ 2M $A_d = g_{m1}R_0 = 35.55$ ⇓ 1M ; $A_{cm} = \frac{32K}{33 \times 120K} = \frac{1}{123.75}$ ⇓ 1M ; $CMRR = 4.4 \times 10^3 \simeq 73dB$ ⇓ 1M $C_m = 51fF$ ⇓ 1M ; $C_L = 38fF$ ⇓ 1M ; $f_{p_1} = \frac{1}{2\pi C_L R_0} = 235.8MHz$ ⇓ 1M ; $f_{p_2} = \frac{g_{m3}}{2\pi C_m} = 3.122GHz$ ⇓ 1M $f_Z(A_d) = \frac{2g_{m3}}{2\pi C_m} = 6.244GHz$; $f_Z(A_{cm}) = \frac{1}{2\pi R_{SS}C_{SS}} = 53MHz$ ⇓ 1M	10M
7(a)	Instrumentation amplifier circuit ⇒ 2M Expression for gain $= \left(1 + \frac{2R_2}{R_1}\right)\frac{R_4}{R_3}$ ⇒ 1M Let $\frac{R_4}{R_3} = \frac{1}{4}$; then $\left(1 + \frac{2R_2}{R_1}\right)$ ⇒ 4 to 320 ; $R_1 = R_{1f} + 50K$ pot ⇒ 1M $\left(1 + \frac{2R_2}{R_{1f}}\right) = 320$ ⇓ 1M and $\left(1 + \frac{2R_2}{R_{1f} + 50K}\right) = 4$ ⇓ 1M ; $R_{1f} = 474.7\Omega$ ⇒ 1M, $R_2 = 75.71K\Omega$ ⇒ 1M	08M
(b)	Difference amplifier circuit ⇒ 2M  Design of 4 resistors ⇒ 1M $\frac{R_2'}{R_1'}$ and $\frac{R_4'}{R_3'}$ worst case values ⇒ 1M ; A_d ⇒ 1M ; A_{cm} ⇒ 2M CMRR ⇒ 1M ; $\frac{R_2'}{R_1'} = 41.6326$ $A_d = 41.592$; $A_{cm} = 0.0812$; $\frac{R_4'}{R_3'} = 38.4313$ $CMRR = 512.2 \simeq 54dB$	08M
8(a)	 Loading effects ⇒ 1M @ output 405K in ll @ input 4.9K in series $\beta = \frac{5}{405}$ ⇒ 1M $A = \frac{v_o}{v_s} = 1200 \times \frac{150}{164.94} \times \frac{7.845}{11.845} = 722.78$ ⇒ 2M $R_i = 164.94K$ ⇓ 1M ; $R_0 = 2.649K$ ⇓ 1M ; $(1 + A\beta) = 9.923$ ⇒ 1M $A_f = 72.84$; $R_{if} = 1.636M\Omega$; $R_{of} = 267\Omega$ } 3M $R_{in} = 1.626M\Omega$; $R_{out} = 276\Omega$	10M
(b)	$P_0 = 4.5W$ ⇒ 1M ; $\eta = 58.9\%$ ⇒ 1M $P_{ai} = 7.64W$ ⇒ 2M ; $P_d = 1.57W$ in each tr ⇒ 1M Minimum Power dissipation capability for each transistor $= \frac{2}{\pi^2} P_0(max) = 1.623$ watts ⇒ 1M	06M

Question No		Marks
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(b)	$g_{m1} = g_{m2} = 2mA/V$; $r_{o1} = r_{o2} = 40K$; $g_{m3} = g_{m4} = 1mA/V$; $r_{o3} = r_{o4} = 32K$; $R_0 = 17.77K$; } $\Rightarrow 2M$ $A_d = g_{m1}R_0 = 35.55 \Downarrow 1M$; $A_{cm} = \frac{32K}{33 \times 120K} = \frac{1}{123.75} \Downarrow 1M$; $CMRR = 4.4 \times 10^3 \simeq 73dB \Downarrow 1M$ $C_m = 51fF \Downarrow 1M$; $C_L = 38fF \Downarrow 1M$; $f_{p1} = \frac{1}{2\pi C_L R_0} = 235.8\,MHz \Downarrow 1M$; $f_{p2} = \frac{g_{m3}}{2\pi C_m} = 3.122\,GHz \Downarrow 1M$ $f_z(A_d) = \frac{2g_{m3}}{2\pi C_m} = 6.244\,GHz$; $f_z(A_{cm}) = \frac{1}{2\pi R_{ss}C_{ss}} = 53\,MHz \Downarrow 1M$	10M
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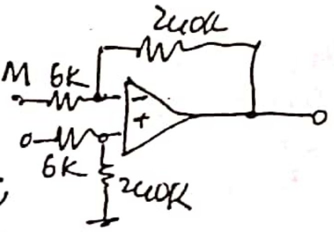
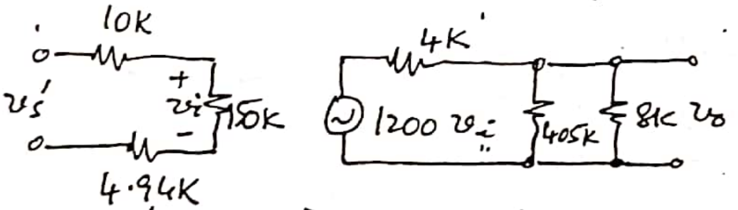
(b)	Difference amplifier circuit $\Rightarrow$ 2M Design of 4 resistors $\Rightarrow$ 1M $\frac{R_2'}{R_1'}$ and $\frac{R_4'}{R_3'}$ worst case values $\Rightarrow$ 1M ; $A_d \Rightarrow 1M$; $A_{cm} \Rightarrow 2M$ $CMRR \Rightarrow 1M$; $A_d = 41.592$; $A_{cm} = 0.0812$; $CMRR = 512.2 \simeq 54dB$  $\frac{R_2'}{R_1'} : 41.6326$ $\frac{R_4'}{R_3'} = 38.4313$	08M
8(a)	 Loading effects $\Rightarrow$ 1M @ output 405K in ll @ input 4.94K in series $\beta = \frac{5}{405} \Rightarrow 1M$ $A = \frac{v_o}{v_s} = 1200 \times \frac{150}{164.94} \times \frac{7.845}{11.845} = 722.78 \Rightarrow 2M$ $R_i = 164.94K \Downarrow 1M$; $R_o = 2.649K \Downarrow 1M$; $(1 + A\beta) = 9.923 \Rightarrow 1M$ $A_f = 72.84$; $R_{if} = 1.636\,M\Omega$; $R_{of} = 267\Omega$ } 3M $R_{in} = 1.626M\Omega$; $R_{out} = 276\Omega$	10M
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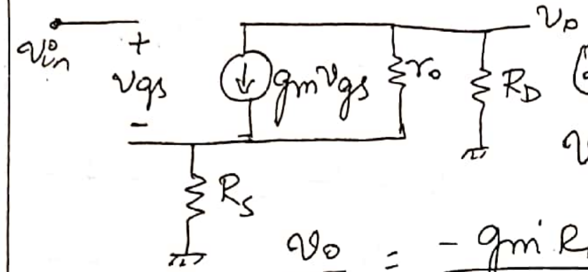
COURSE CODE: 16EC33 **COURSE:** Analog Microelectronic Circuits UG

Question No		Marks

PART - A

1.1. 100 → 1 ; 1.2. 4 → 1 ; 1.3. 1KΩ → 1 ; 1.4 → 1mA → 1 ;
1.5. 5V → 1 1mA → 1 ; 1.6. 11KΩ → 1 9.09Ω → 1 ;
1.7. 0.998mA → 1 ; 1.8. Obtain small currents with small resistor values → 1 ; 1.9. 10 → 2 ; 1.10. 12mA → 2
1.11. 8MHz → 1 ; 1.12. 50V/ms → 1 ; 1.13. 10MΩ → 1
1Ω → 1 ; 1.14. increases, decreases → 1 ; 1.15. 24W → 1 — **20**

PART - B

2a. 

(2)

Converting current source to voltage source → 1

$$v_o = -\frac{g_m v_{gs} r_o}{r_o + R_D + R_S} R_D \rightarrow 1$$

$$\frac{v_o}{v_{in}} = \frac{-g_m R_D}{1 + g_m R_S + \frac{(R_D + R_S)}{r_o}} \longrightarrow 2$$

Output impedance small signal equivalent → 2

$V_x + g_m V_1 r_o = I_x (R_S + r_o)$, — 1 $R_{out} = R_S + r_o + R_S g_m r_o$ → 1 — **10**

2b. $V_2 = 6V$; → 1 Let $I_D = x\,mA$; $2x = 2(6 - 2x - 1)^2$ — 2

$x = 2mA$ → 1 $V_3 = 2V$ 4V → 1 $V_1 = 10V$ 8V → 1 — **6**

3a. $\frac{v_o}{v_{in}} = \frac{R_C \| R_L}{r_e}$; $r_e = 26\Omega$ → 1 $R_C = 7.43K\Omega$ → 1 ;

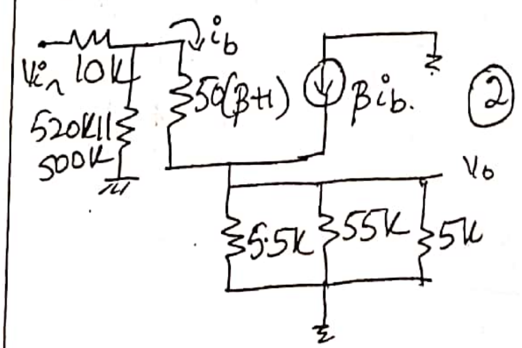
$S(I_{CO}) = \frac{(\beta+1)(R_B + R_E)}{R_B + (\beta+1)R_E}$ → 0.5

$18 = 7.35 + 6 + V_E$ → $R_E = 4.64K\Omega$ → 1

$R_B = 46.4K\Omega$; → 0.5 $V_{TH} = 5.8$ → 1 ; $R_1 = 144K\Omega$ → 1

$R_2 = 68.46K\Omega$ → 1 $R_{in} = 2.462K\Omega$ → 1 — **8**

3b. $r_e = 50\Omega$; $R_{in} = 520K \| 550 \| (\beta+1)[50 + (5.5K \| 5K \| 55K)] = 127.5K\Omega$ (2)



(2)

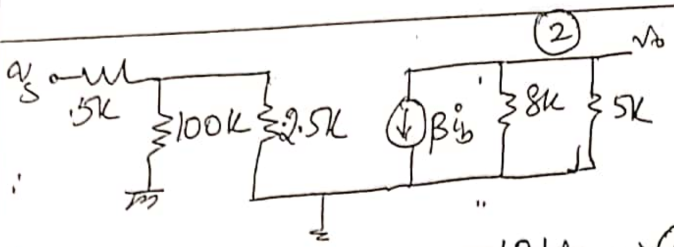
$R_{out} = 5.5K \| 55K \| [(\beta+1)50 + (10K \| 520K \| 500K)]$

$= 142\Omega$ → 2

$$\frac{V_o}{V_{in}} = \frac{2.5K}{2.5K + 50} \times \frac{127.5K}{127.5K + 10K} = 0.91 \rightarrow 2$$ — **8**

Question No		Marks

4a



$$\frac{v_o}{v_s} = \frac{-8K\|5K}{25} \times \frac{100K\|2.5K}{5K+(100K\|2.5K)}$$

$$= 123.07 \times 0.328 = -40.35 \quad (2)$$

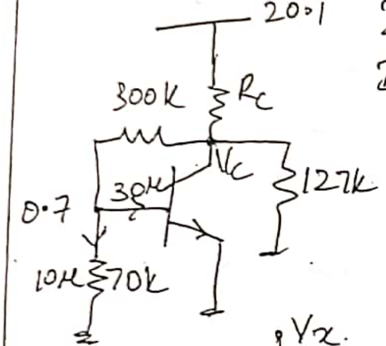
$f_{C_{C1}} = \dfrac{1}{2\pi C_1[5K+(100K\|2.5K)]} = 10\,Hz \rightarrow (1)$ $\quad f_{C_{C2}} = \dfrac{1}{2\pi C_2[5K+8K]} = 10\,Hz \rightarrow (1)$ (10)

$f_{CE} = \dfrac{1}{2\pi C_E\left[R_E \| \left[r_e + \frac{R_B\|5K}{100}\right]\right]} = 70.9\,Hz \rightarrow (1)$ $\quad C_{in} = C_{be} + (1+A_v)C_{bc}$

$= 130.95\,pf - (1)$

$f_{h_i} = \dfrac{1}{2\pi C_{in} R_{in}} : 741\,KHz \rightarrow (1)$ $\quad f_{h_o} = \dfrac{1}{2\pi C_o R_o} = 51.3\,MHz \rightarrow (1)$

4b.



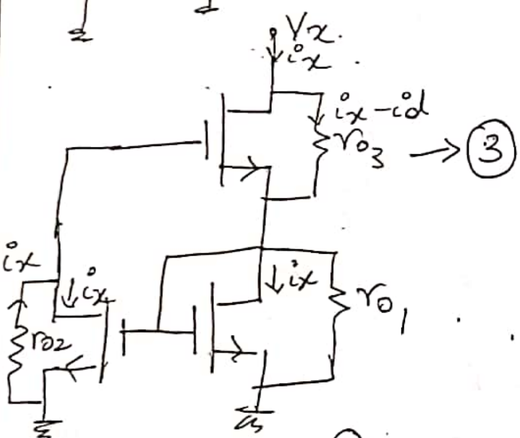
I through 300K = 40µA → (1) $\quad V_C = 12.7 \rightarrow (1)$

I through 127K = 100µA → (1) $\quad I_C = 600\mu A \rightarrow (1)$ (6)

I through R_C = 740µA → (1) $\quad R_C = \dfrac{20.1-12.7}{740\mu}$

$= 10K\Omega \rightarrow (1)$

5a.



$i_d = g_{m3} v_{gs3} - (1)$

$i_d = g_{m3}\left[i_x \cdot r_{o2} - \dfrac{i_x}{g_{m1}}\right] - (1)$ (6)

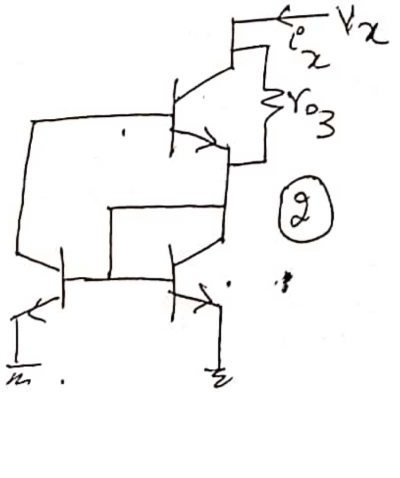
$\dfrac{v_x}{i_x} \simeq g_{m3} r_{o2} r_{o3} - (1)$

5b. $g_{mN} = 2mA/V - (1)$ $\quad g_{mp} = 1mA/V - (1)$ $\quad r_{op} = r_{oN} = 16K\Omega - (1)$ $\quad A_d = 16 \rightarrow (1)$ (10)

$A_{CM} = -5m - (0.5)$ $\quad CMRR = 70dB \rightarrow (0.5)$; $C_M = 65pf$ (0.5); $C_L = 13pf \rightarrow (0.5)$

$f_{p_1} = 24.48MHz \rightarrow (1)$ $\quad f_z = 48.97MHz - (1)$ $\quad f_{p_2} = 1.53MHz \rightarrow (1)$ $\quad f_{CM} = 15.91KHz - (1)$

6a



$i_x = 2\left(1+\dfrac{1}{\beta}\right) = (2)$

$V_x = i\left[2+\dfrac{2}{\beta}+\beta\right] r_{o3} \rightarrow (1)$ (6)

$\dfrac{V_x}{i_x} \approx \dfrac{\beta r_{o3}}{2} \rightarrow (1)$

SUBJECTCODE: SUBJECT: UG

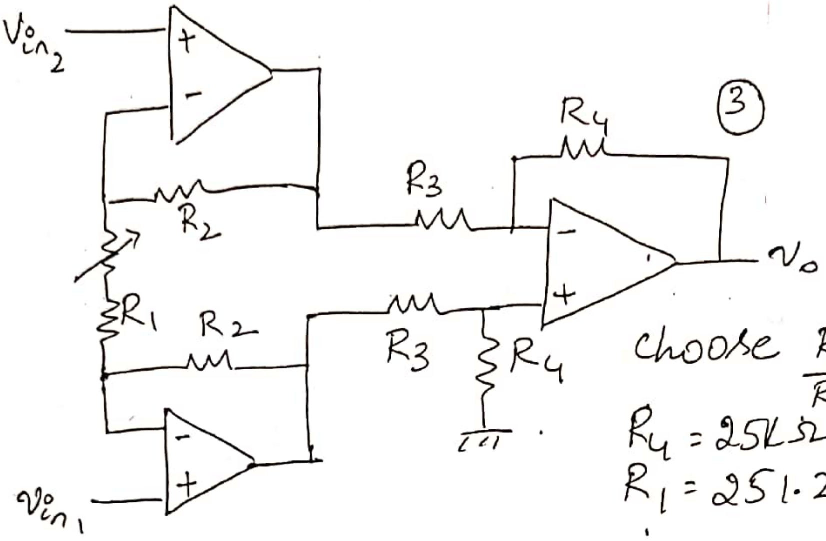
Question No		Marks

6b. $r_e = 26\Omega \rightarrow 1$ $R_{id} = 2(\beta+1)(r_e + R_E) \doteq 20K\Omega \rightarrow 2$ $R_{icm} = (\beta+1)(R_{EE} \| \frac{r_o}{2})$ $= 5M\Omega \rightarrow 2$ — **10**

$A_d = \frac{10K \| 200K}{74 + 26} = 95.23 \rightarrow 2$ $A_{CM} = 0.5m \rightarrow 2$

$CMRR = 105.6\,dB.$

7a.



3

$0.5 = \frac{R_4}{R_3}\left[1 + \frac{2R_2}{R_1 + 100K}\right] \rightarrow 1$

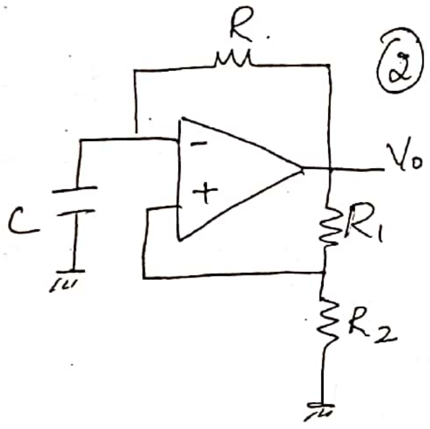
$100 = \frac{R_4}{R_3}\left[1 + \frac{2R_2}{R_1}\right] \rightarrow 1$ — **8**

choose $\frac{R_4}{R_3} = 0.25$

$R_4 = 25K\Omega$ $R_3 = 100K\Omega - 1$

$R_1 = 251.2\Omega - 1$ $R_2 = 50.125K\Omega$ 1

7b.



2

$T = 10\mu s \rightarrow 1$

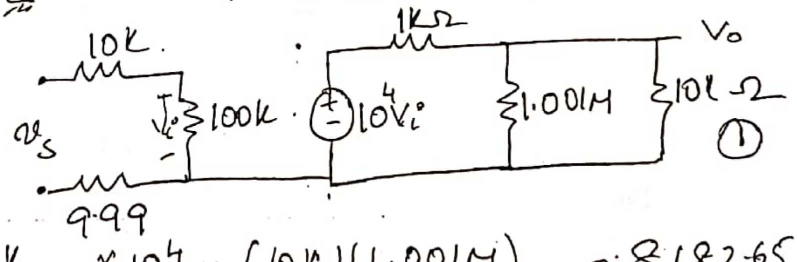
$T = RC\ln\left(\frac{1+\beta}{1-\beta}\right)$ where $\beta = \frac{R_2}{R_1 + R_2}$ 1 — **8**

Let $R = 1K\Omega$; $C = 0.01\mu f - 1$

$1 = \ln\left(\frac{1+\beta}{1-\beta}\right) \Rightarrow \beta = 0.462 - 1$

$R_1 = 5.38K\Omega \rightarrow 1$, $R_2 = 4.62K\Omega \rightarrow 1$

8a $\beta = 0.999m$ 0.5



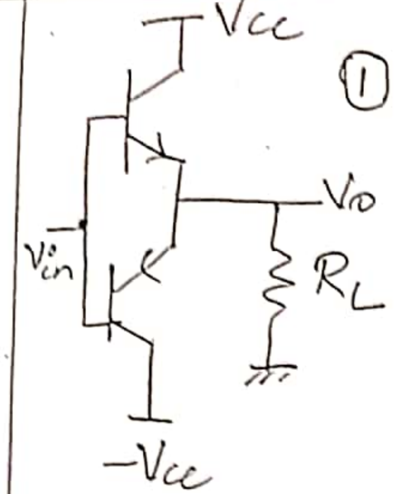
1

$A = \frac{V_o}{V_s} = \frac{100K}{10K + 100K + 999} \times 10^4 \times \frac{(10K \| 1.001M)}{1K + (10K \| 1.001M)} = 8182.65 \rightarrow 1$ — **8**

$R_{in} = 110.999K\Omega \rightarrow 0.5$. $R_{out} = 908.26\Omega \rightarrow 0.5$

$1 + A_v\beta = 9.174 \rightarrow 0.5$ $A_f = 891.839 \rightarrow 1$

$R_{if} = 1.008M\Omega \rightarrow 1.5$ $R_{of} = 99.99\Omega \rightarrow 1.5$

Question No		Marks
8b	 1 $P_{ac} = \frac{V_{cc}}{\sqrt{2}} \cdot \frac{I_L(p)}{\sqrt{2}} \rightarrow 1$ $P_{dc} = V_{cc}\, I_L(p) \times \frac{2}{\pi} - 1$ $\eta = \frac{P_{ac}}{P_{dc}} = 78\% - 1$	4
8c.	$P = \frac{T_{j(max)} - T_A}{\theta_{cs} + \theta_{SA} + \theta_{jc}} = \frac{200-25}{0.5+1.5+1.0} = 43.75W \longrightarrow 2$ $T_{HS} = (1.5)(43.75) + 25 = 90.625^\circ C - 1$ $T_C = 200 - (1)(43.75) = 156.25^\circ C - 1$	4

<—— End ——>

$\cancel{6} - V_D = \cancel{6} - 3 - 1$

$V_D = 4.$

$-g_m$

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R. V. COLLEGE OF ENGINEERING

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III / IV Semester B. E. Fast Track Examinations July-18

Electronics and Communication Engineering

ANALOG MICROELECTRONIC CIRCUITS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1

1.1 $NMOS$ transistor is operated with an overdrive voltage $V_{OV} = 1\,V$ and has an early voltage $V_A = 50\,V$. Its intrinsic gain is ________. **01**

1.2 $PMOS$ is operated with $V_G = 3V, V_s = 6V$ and $V_T = |1V|$. V_D required to operate $PMOS$ at the edge of saturation is ________. **01**

1.3 For the circuit shown in fig 1.3, calculate the incremental impedance between A and B. Assume $NMOS$ is operating with $I_D = 1\,mA$, $\lambda = 0$ and $K_n'\left(\frac{W}{L}\right) = 0.5\,mA/V^2$. **01**

1.4 For the circuit shown in fig 1.4, the drain current $I_D =$ ________. Assume $K_n'\left(\frac{W}{L}\right) = 2\,mA/V^2$ and $V_t = 1V$. **01**

1.5 The operating point, V_{CE} and I_C, for the circuit shown in fig 1.5 are ________ and ________.

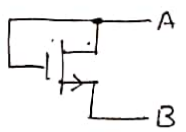


Fig 1.3

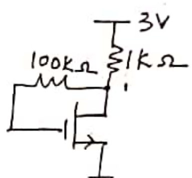


Fig 1.4

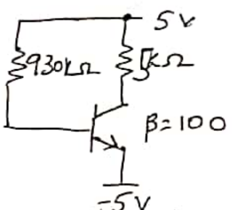


Fig 1.5

02

1.6 The input impedance R_{in} and output impedance R_{out} for the circuit shown in fig 1.6 are ________ and ________ respectively. Assume $\beta = 99, \lambda = 0$ and $I_E = 2.6\,mA$. **02**

1.7 In the circuit shown in figure 1.7, the output current $I_0 =$ ________. Assume all transistors are identical having $\beta = 40$.

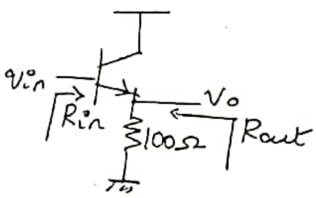


Fig 1.6

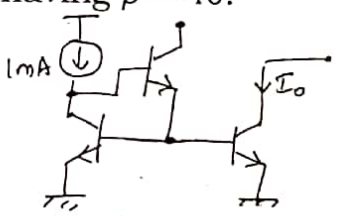
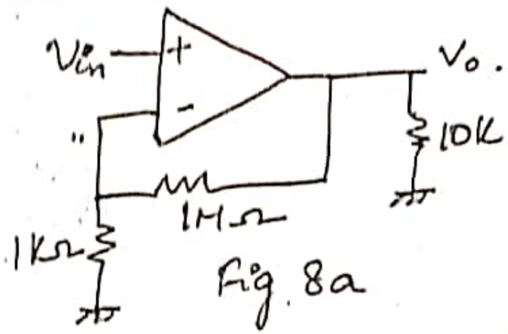


Fig 1.7

01

1.8 Mention the main advantage of Widlar current mirror. **01**

1.9 The voltage gain (V_0/V_{id}) for the circuit shown in fig 1.9 is ________. Assume both $PMOS$ and $NMOS$ transistors are identical. **02**

6	a	Derive an expression for the output impedance of Wilson mirror using *BJTS*.	06
	b	For the circuit shown in fig 6b, determine re, differential input impedance Rid, common mode input impedance R_{icm}, differential voltage gain $\frac{v_0}{v_{id}}$, common mode voltage gain V_o/V_{icm} and $CMRR$ for one percent mismatch in R_c.	10
7	a	Design an instrumental amplifier to realize a differential gain, variable in the range 0.5 to 100, utilizing a 100 k pot as a variable resistor.	08
	b	Design an astable multi-vibrator to obtain a square wave of frequency 50 kHz.	08
8	a	In the amplifier circuit of fig 8a, determine V_o/V_s, R_{in} and R_{out} using feedback methodology. The Op-Amp has an open loop gain 10^4, open loop input and output impedance 100 $k\Omega$ and 1 $k\Omega$ respectively.  Fig. 8a	08
	b	Prove that the maximum conversion efficiency of class B complementary symmetry amplifier is 78%.	04
	c	A silicon power transistor operated with heat sink $[\theta_{cs} = 1.5\ C/W, \theta_{SA} = 1.5°C/W]$ has $\theta_{jc} = 1°C/W$. What is the maximum power that could be dissipated by the transistor, if $T_j(max) = 200°C$ at $T_A = 25°C$? Also calculate the heat sink temperature and case temperature.	04

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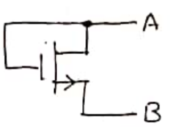
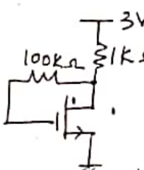
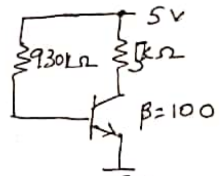
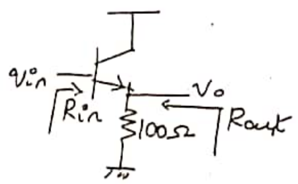
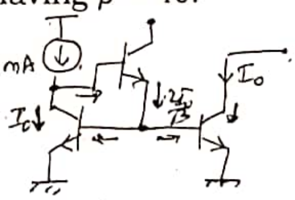
PART-A

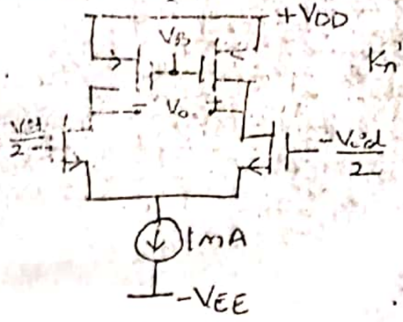
1	1.1	$NMOS$ transistor is operated with an overdrive voltage $V_{OV} = 1\,V$ and has an early voltage $V_A = 50\,V$. Its intrinsic gain is 100.	01
	1.2	$PMOS$ is operated with $V_G = 3V, V_s = 6V$ and $V_T = \|1V\|$. V_D required to operate $PMOS$ at the edge of saturation is 4.	01
	1.3	For the circuit shown in fig 1.3, calculate the incremental impedance between A and B. Assume $NMOS$ is operating with $I_D = 1\,mA$, $\lambda = 0$ and $K_n'\left(\frac{W}{L}\right) = 0.5\,mA/V^2$. 1KΩ	01
	1.4	For the circuit shown in fig 1.4, the drain current $I_D =$ 1mA. Assume $K_n'\left(\frac{W}{L}\right) = 2\,mA/V^2$and $V_t = 1V$.	01
	1.5	The operating point, V_{CE} and I_C, for the circuit shown in fig 1.5 are 5V and 1mA.	
		 Fig 1.3  Fig 1.4  Fig 1.5	02
	1.6	The input impedance R_{in} and output impedance R_{out} for the circuit shown in fig 1.6 are 11K and 9.09 respectively. Assume $\beta = 99, \lambda = 0$ and $I_E = 2.6\,mA$.	02
	1.7	In the circuit shown in figure 1.7, the output current $I_0 =$ 0.998mA. Assume all transistors are identical having $\beta = 40$.	
		 Fig 1.6  Fig 1.7	01
	1.8	Mention the main advantage of Widlar current mirror. small I for small R	01
	1.9	The voltage gain (V_0/V_{id}) for the circuit shown in fig 1.9 is 10. Assume both $PMOS$ and $NMOS$ transistors are identical.	02

$Z_i = (\beta+1)(r_e+R_E) = 11.1K$

$Z_o = r_e \| R_E = 9.09$

$I_0 = \frac{I_{ref}}{1+\frac{2}{\beta^2}} = 0.9987mA$

1.10 The current I_0 for the circuit shown in fig 1.10 is 12mA. Assume ideal op-amp.



$K_n'\left(\frac{W}{L}\right) = 1mA/V^2$

$V_{AN} = |V_{AP}| = 10V$

Fig 1.9

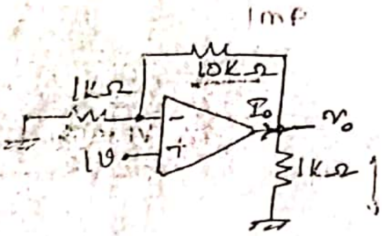


Fig 1.10

$V_0 = 0$

1K.

11mA

02 20

1.11 An op-amp has unity gain bandwidth of 40 MHz. If it is used as an amplifier with closed loop gain of 5, the bandwidth of closed loop amplifier is 8MHz . 01

1.12 An Op-Amp connected as voltage follower has the output voltage $V_0 = 5\left[1 - e^{-t(10k)}\right]$. Its slew rate is 50V/ms 01

1.13 An Op-Amp has open loop gain 999, input impedance 10 $k\Omega$ and output impedance 1 $k\Omega$. If, the Op-Amp is connected as voltage follower, then the input and output impedances of voltage follower are 10MΩ and 1Ω respectively. 02 13

1.14 Voltage series negative feedback is used to ↑ the input impedance and ↓ the output impedance of the amplifier. 01

1.15 A dual supply of 12 V is used in a complementary symmetry class B power amplifier with load 8Ω. The maximum power output possible is ~~2W~~. 01

$v_o = -\frac{g_m V_{gs} r_o' R_D}{r_o + R_D + R_S}$ $\quad \frac{v_o}{v_{in}} = -\frac{g_m R_D}{1 + g_m R_S + \left(\frac{R_D + R_S}{r_o}\right)}$ $\quad R_{out} = R_S + r_o + R_S g_m r_o$

PART-B

2 a Draw the small signal equivalent of common source amplifier with degeneration resistance and derive the expression for voltage gain and output impedance. Assume $\lambda \neq 0$. 10

8V, 6V 4V

b In the circuit of figure 2b, determine V_1, V_2 and V_3. Q_1 and Q_2 are identical with $K_n'\left(\frac{W}{L}\right) = 4\ mA/V^2$ and $V_t = 1V$. 06

3 a Design a common emitter amplifier circuit shown in fig 3a with $V_{CE} = 6V$ and $I_E = 1\ mA$ with $S(I_{CO}) = 10$ and small signal voltage gain $V_0/V_i = -100$. For the designed circuit determine the input impedance. Assume $\beta = 99$ and $\lambda = 0$. 08

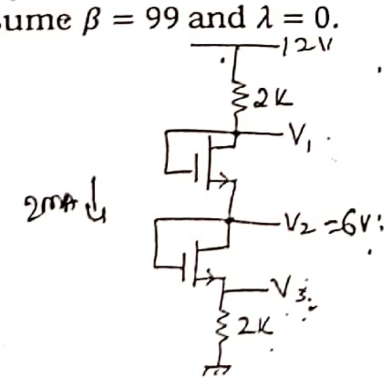


Fig 2b

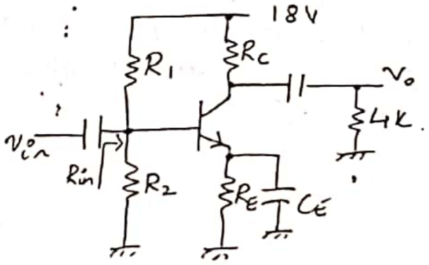


Fig 3a

$\frac{V_o}{V_{in}} = \frac{R_C \| R_L}{r_e}$ $\quad r_e = 26$

$R_C = 7.43 k\Omega$

$R_E = 4.64 k\Omega$

$R_B = 46.4 k\Omega$

$R_1 = 144K$

$R_2 = 68.46K$

$R_{in} = 2.462 k\Omega$